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THE AERODYNAMIC DESIGN OF BLOWER TUNNELS WITH WIDE-ANGLE DIFFUSERS

R. D. Mehta

Department of Aeronautics, Imperial College, London

Summary—The aerodynamic design of open-circuit wind tunnels, driven by a centrifugal "blower" connected to the settling chamber by a wide-angle diffuser, is discussed. The design or choice of the blower, diffuser and screens is described in detail because of the strong influence of these components on tunnel performance. Contraction and working section designs, which have been studied widely over the years, differ little between open-circuit and closed-circuit tunnels, and are thus discussed here in lesser detail. Some advice on structural considerations and their influence on the aerodynamic design of tunnels is also included. The design rules and suggestions, based mainly on data collected from successful blower tunnel designs, are supplemented by some new experimental results for wide-angle diffusers, screens and blowers.

LIST OF SYMBOLS

A		area or diffuser area ratio
c		contraction ratio
C_f		skin friction coefficient
C_p	$= \frac{P - P_{at}}{\frac{1}{2}\rho U_{in}^2}$	local total pressure coefficient
d		screen wire diameter
F		force coefficient parallel to the gauze
H		power input or boundary layer shape parameter
K	$= \frac{\Delta p}{q}$	screen pressure drop coefficient
K_{sum}	$= \Sigma\frac{\Delta p}{q}$	sum of pressure drop coefficients of all the screens in a diffuser
L		diffuser length
ℓ		screen mesh length
n		number of screens within a diffuser
O.P.	$= \frac{\Delta p}{\frac{1}{2}\rho U_{ex}^2}$	blower operating point
P		total pressure
p		static pressure
Δp		static pressure drop across screen or fan static-pressure rise
Q		volume flow rate through blower

q = $\frac{1}{2}\rho U^2$, dynamic pressure

r_1 inner radius of rotor

r_2 outer radius of rotor

Re Reynolds number

U, V, W mean velocity components (U-axial)

u, v, w fluctuating velocity components (u-axial)

U_{ex} = $\frac{Q}{A_{out}}$ average mean velocity of blower exit flow

x axial distance in streamwise direction

Greek symbols

α = $\frac{\phi}{\theta}$, deflection coefficient of screen

β = $(1 - \frac{d}{\ell})^2$, screen open-area ratio

δ boundary layer thickness

δ^* boundary layer displacement thickness

η = $\frac{P_{out} - P_{in}}{\frac{1}{2}\rho(U_{in}^2 - U_{out}^2)}$

θ boundary layer momentum thickness or angle between incident flow and local normal to the screen or half the diffuser angle

λ = $\frac{H}{\frac{1}{2}\rho_0 U_0^2 A_0}$, tunnel power factor

μ viscosity or = $\frac{1}{\alpha}$, screen refractive index

ν = $\frac{\mu}{\rho}$, kinematic viscosity

ρ density

ϕ angle between emerging flow and local normal to the screen

ω fan rotational speed, rad/s

Subscripts

o working section

at atmospheric

in inlet

out outlet

θ screen inclined to the incident flow.

1. INTRODUCTION

The design of a blower-driven wind tunnel such as that shown in Fig. 1 is a combination of art, science and common sense, the last being the most essential. It is difficult and unwise to predict firm rules for tunnel design. This is mainly due to the wide variety of tunnel designs and the lack of understanding of flow through wind tunnel components like the diffuser, screens and the blower itself. It is more feasible and sensible to predict approximate design boundaries based on data from existing tunnels which are known to perform satisfactorily. This is what is attempted in this review. Some original experiments, on what are thought to be the more important components, are also described. There has been no comprehensive guide to wind tunnel design since the account by Bradshaw and Pankhurst (1964),* and this review is intended to be an updated and extended version of a part of that paper, of which some knowledge will be assumed.

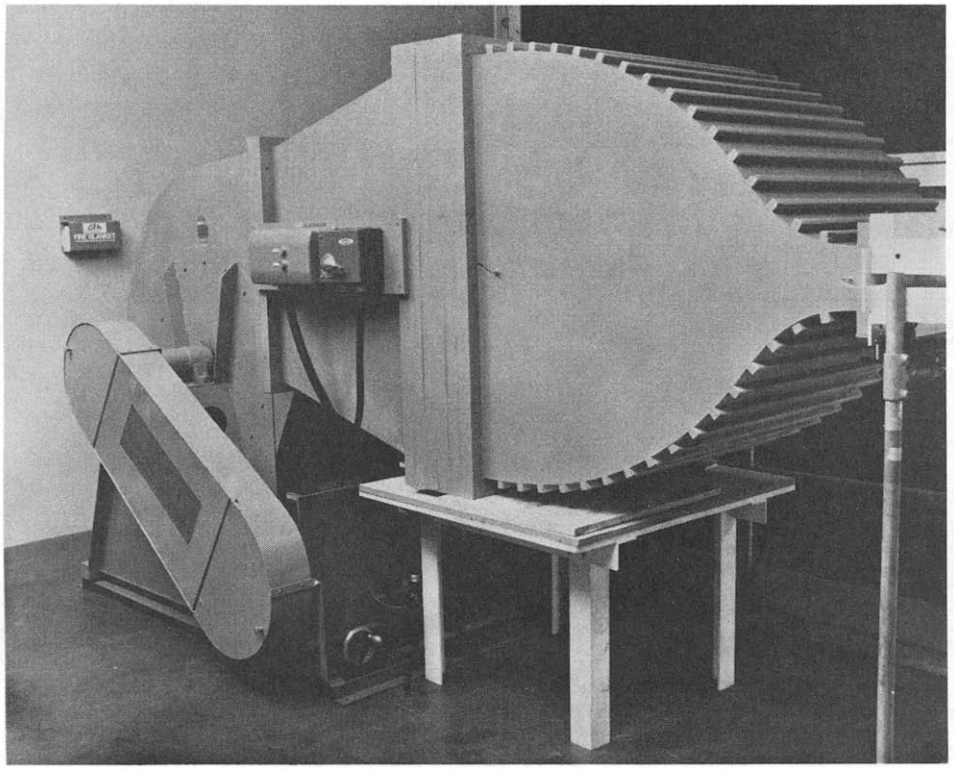


Fig. 1a. A typical blower tunnel (modified version of the tunnel at Imperial College described by Bradshaw, 1972).

The earliest wind tunnels were open-circuit suction designs with an unobstructed intake and an axial fan at the exit (Fig. 2) although the Wright Brothers' tunnel had an axial fan in the intake. The expanding passage or "diffuser" downstream of the working section is intended to slow the flow down and recover some of its kinetic energy in the form of a pressure rise, thus reducing power consumption. Some sort of transition section between the working section and the fan is essential unless the working section is circular, and the optimum fan efficiency is obtained if the cross-sectional area at the fan is between 2 and 3 times that of the working section. It is therefore usual to place the fan near (often at) the end of the diffuser.

*Copies may be obtained from the Microform division of Pergamon Press, or, at cost, as Xerox copies from P. Bradshaw at Imperial College.

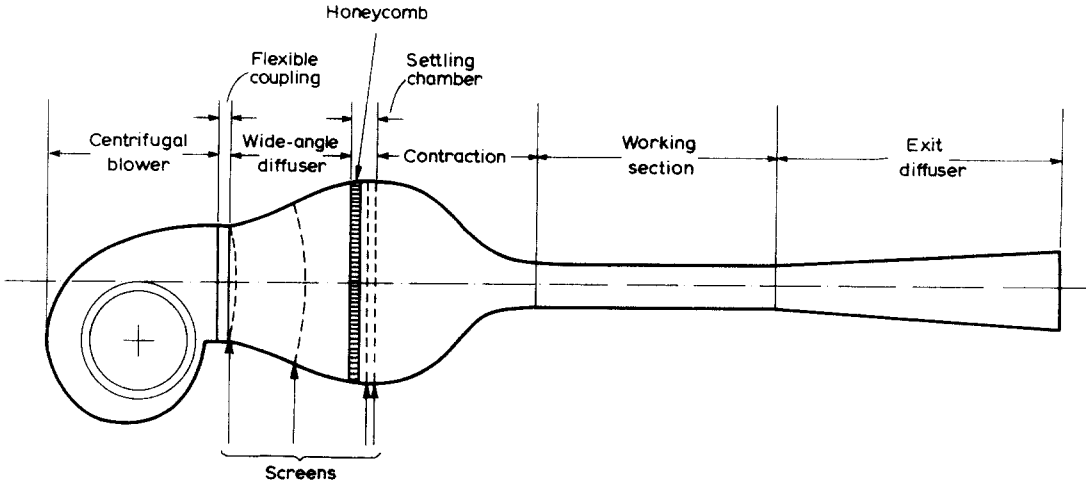


Fig. 1b. The main components of a blower tunnel. (Not to scale).

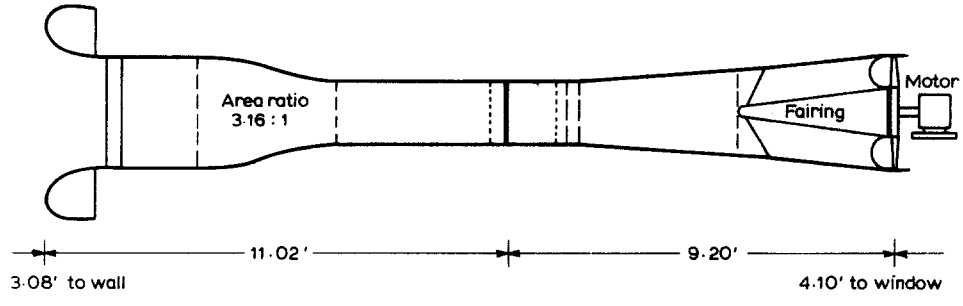


Fig. 2. NPL 10 in. suction tunnel (1947).

Since the 1930s, when the strong effect of free-stream turbulence in triggering early transition of the laminar boundary layer became apparent, emphasis has been laid on wind tunnels with low turbulence levels. Consequently, with very few exceptions, wind tunnels too large for a laboratory, and many laboratory-size tunnels as well, have been designed as closed-circuit types to ensure a smooth return flow. However, it has been found that open-circuit blower tunnels with centrifugal blowers (with backward-facing aerofoil-type blades) at the entry (Fig. 1), are much less sensitive to disturbances in the entry flow than suction tunnels. Also, centrifugal blowers run with reasonable steadiness and efficiency over a much wider range of flow conditions than axial fans do, so that quite large changes in power factor can be accepted. Furthermore disturbances or blockage in the working section can greatly affect the flow into a fan downstream but have virtually no effect on an upstream blower. Some of the first blower tunnels designed include: Clauser's boundary layer tunnel (1954) and the Liverpool University 4' × 2' (1959), although the latter had an axial fan.

The general arrangement of a blower tunnel is similar to that of a suction tunnel. However, a diffuser is generally needed between the blower and the settling chamber, while the diffuser downstream of the working section is no longer necessary for the fan's sake. It may even be omitted if flexibility of working-section arrangements is more important than pressure recovery Pressure recovery in the diffuser that matches the blower to the settling chamber is usually unimportant and it is now customary for this diffuser to be a rapid ("wide-angle") one with some means of boundary layer control. Such diffusers have also been used for some time past

in closed-circuit tunnels, usually between the fourth corner and the settling chamber: McLellan and Nichols (1942), Squire and Hogg (1944), RAE 4' × 3' (1946) (Squire and Winter, 1953) and Cambridge University cascade tunnel (Rhoden, 1951). In both cases the object is to reduce the overall length of the tunnel. The details of most of these tunnels, and many more, can be found in Bradshaw and Pankhurst (1964).

The main aim when designing any wind tunnel is to produce a steady flow with spatial uniformity in the working section, over a range of Reynolds numbers. The Reynolds number (Re), which is proportional to the ratio of a typical inertia force to a typical viscous force and strongly influences the behaviour of boundary layers and wakes, is the most important non-dimensional group as far as low speed wind tunnel design is concerned. It is not essential to reproduce the exact Re in model tests since transition of boundary layers can be artificially stimulated and quantities like the skin friction, whose variation with Re is known, modified accordingly. For a given Re, the required speed of the flow is inversely proportional to the model size. The upper speed limit for blower tunnels is set by the blower power and structural limitations. The power consumption of a tunnel increases as $(\text{speed})^3$ and loads on the tunnel structure and model increase as $(\text{speed})^2$. The annoyance value of draughts in the room also increases as $(\text{speed})^2$: a working section Re per metre of more than about 3×10^6 (a speed of about 40 ms^{-1}) is rare in blower tunnels of whatever size, and commercial blowers capable of producing such a speed in a working section more than about 1 m^2 in area are also rare.

The configuration of a wind tunnel is often decided by the size of the working section. Most tunnels with a working section diameter greater than about a metre have return circuits with cascades of corner vanes. (The term "diameter" is defined as the diameter of a circle with the same cross-sectional area as the component). One of the main reasons for this is that a large open-circuit tunnel would be of rather inconvenient dimensions, mainly in length; in a return-circuit tunnel, the exit diffuser can accomodate the first corner. This problem is further magnified by the requirement that the open-tunnel axis must not be too near the ceiling, floor or side walls, and the inlet must not be closer than 1 - 1½ times its own diameter from the end wall. This is to ensure that the "open-return" circuit through the laboratory is not unduly interfered with. (It is also known for blowers placed with their inlet too near an obstruction to develop a "bath-tub" vortex at the entry which results in the blower behaving abnormally.) In some suction tunnel installations a honeycomb or screen partition has to be fitted across the room, to ensure that the return flow arrives at the entry with a relatively small degree of swirl. On the other hand, a room containing an open-circuit tunnel should not be too large since the performance of these tunnels is liable to suffer from draughts. The other principal reason why large open-circuit tunnels are unpopular is that, on average, closed-circuit tunnels have a better flow and lower power factor $\left(\lambda = \frac{H}{\frac{1}{2}\rho_o U_o^3 A_o}\right.$, defined as the ratio of input power to the rate of kinetic energy flux in the working section$\left.\right)$ than the average open-circuit tunnel. Since the disturbed air from the working section is not directly recirculated in an open-circuit tunnel, and because of the absence of corner vanes, one would tend to believe the converse, and indeed the Indian Institute of Science at Bangalore has built a 14' × 9' (4.3 m × 2.7 m) open-circuit tunnel with a maximum speed of 350 ft/s (100 m/s) and a power factor of about 0.15.

Blower tunnels are often used for cascade tests, boundary layer studies and similar work where the flow leaving the working section is greatly disturbed. The omission of an exit diffuser necessarily makes the power factor, λ, greater than unity because the whole of the kinetic energy flux through the working section is lost at exit, but the gain in flexibility of working

section arrangement usually outweighs this. Some of the blower tunnel working sections used in the Aeronautics Department at Imperial College are shown in Fig. 3. Even with such contrasting working section arrangements, a high-quality flow can be obtained in a well designed blower tunnel. This, together with the saving of space and constructional costs makes the blower tunnel an attractive proposition both for general purpose work and for specialized working section arrangements.

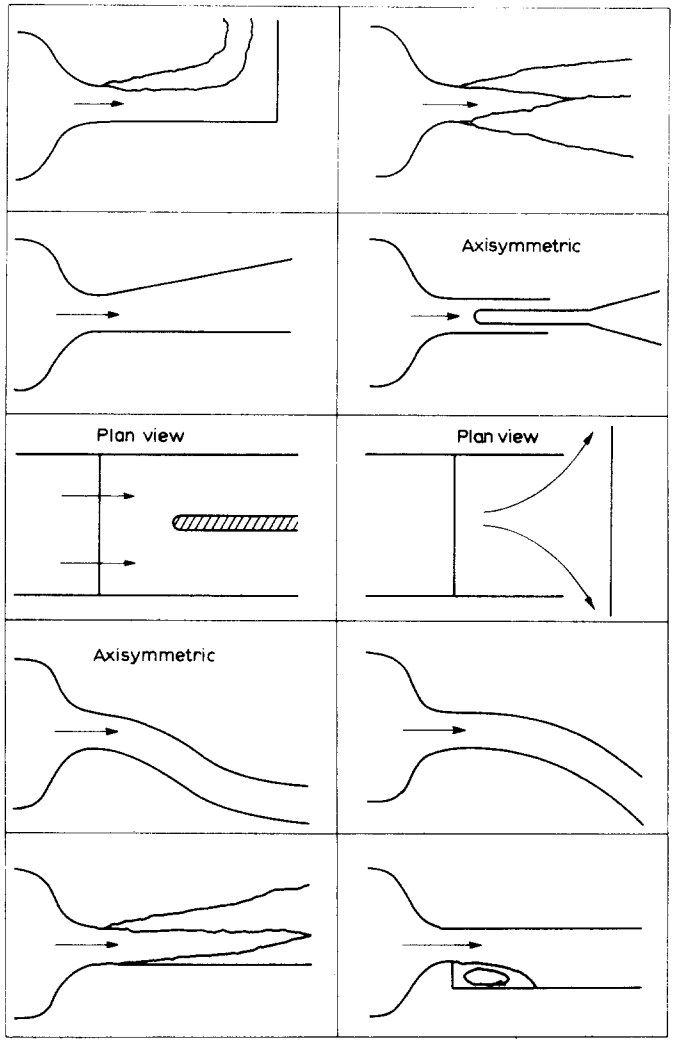


Fig. 3. Blower tunnel configurations used at Imperial College.
(Wavy lines are shear-layer edges)

The main components of a blower tunnel, starting from the upstream end, are the centrifugal blower, wide-angle diffuser, settling chamber, contraction, working section and sometimes an exit diffuser (Fig. 1).

The air is essentially "sucked" through the centrifugal blower inlet and a total-pressure rise imposed across the rotor. It is important to assess the quality of the flow from a centrifugal blower and how it changes, because the performance of components downstream is dependent on it. In the wide-angle diffuser the cross-section is expanded, usually in the least possible axial distance, subject to the avoidance of severe flow separation. Separation here usually leads to unsteadiness in the working section flow. The most popular means of preventing severe separation in the diffuser is to install wire-gauze screens. Screens operate by reducing the

total pressure of high-speed regions more than that of low-speed regions (e.g. boundary layers) so that downstream of the region of static pressure perturbations caused by the screens, the velocity is more uniform and the boundary layer further from separation. The screens in the wide-angle diffuser incidentally help to reduce the turbulence of the flow, but further screens and the honeycomb, intended primarily to reduce turbulence, are installed in the settling chamber, where the cross-sectional area is largest, so as to minimize the losses. The honeycomb reduces irregularities in the flow direction, removing any swirl still present, and the screens improve the flow uniformity and steadiness besides further reducing the turbulence intensity. In the contraction, the mean velocity is increased and percentage spatial or temporal variations are further reduced. This flow, with the desired quality, then passes through the working section. The exit diffuser option is settled by considering the tunnel power factor and working section flexibility. The order in which these components are discussed below reflects the relative importance attached to them as far as the tunnel performance is concerned, rather than a recommended sequence of design.

We start the detailed discussion of blower-tunnel design with an account of the design and performance of wide-angle diffusers (Section 2) followed by an analysis of the flow through gauze screens (Section 3) and of the aerodynamic performance of centrifugal blowers with special reference to the exit flow (Section 4). The design of the remaining tunnel components is much the same as for return-circuit tunnels and is discussed in Section 5. Design rules and suggestions are summarised in Section 6. Each section is nearly self-contained, vital information from other sections being repeated for convenience.

2. WIDE-ANGLE DIFFUSERS

2.1. General considerations

Although diffusers are widely used, their flow characteristics are still not fully understood. The flow through a diffuser inevitably depends on its geometry, defined by the area ratio, wall expansion angle, cross-sectional shape and wall contour. Other parameters such as conditions at entry and exit and boundary layer control devices, if any, also affect the diffuser performance. With an arbitrary combination of these parameters, the flow through a diffuser becomes too difficult to predict in detail. The issue is further complicated by the occasional presence of flow separation caused by the adverse pressure gradients present in a diffuser.

These features coupled with the general problems of predicting the behaviour of turbulent flow make the diffuser one of the least understood fluid flow devices. Almost all the knowledge acquired about diffusers is empirical.

Diffusers can be classified into two distinct categories:

(i) Exit diffusers. Exit diffusers, as fitted, for example, after a wind tunnel working section, generally have very gentle expansions with an "equivalent cone angle" of between about 5° (for best flow steadiness) and 10° (for best pressure recovery), and an area ratio (outlet to inlet) not exceeding about 2.5. The equivalent cone angle of a diffuser is defined as the included angle of a circular cone having the same inlet area, outlet area and length as the given diffuser. In the case of a two-dimensional diffuser, the concept of an equivalent wedge is more useful. It is important to have a reasonable degree of flow steadiness in the exit diffuser, since otherwise the pressure recovery tends to fluctuate with time, and therefore so does the tunnel speed if the input power is nearly constant. The flow in this class of diffusers is intended to be fully attached and their design is fairly well catered for by existing methods. For a review, see Cockrell and Markland (1974).

(ii) Wide-angle diffusers. The uniformity of flow in a wind tunnel can be greatly improved if a contraction section or nozzle of large area ratio is provided immediately upstream of the working section. If a large diffuser area ratio is required, so as to provide a large contraction ratio, a long conventional diffuser, with an equivalent cone angle of 5°, is undesirable economically. For instance if a modest contraction ratio of 12 is required a 5° diffuser with a length of 20 times the working-section diameter would be needed. A short diffuser, with a large area ratio and consequently a large equivalent cone angle, is then installed. This is a "wide-angle" diffuser, defined as a diffuser in which the cross-sectional area increases so rapidly that separation can be avoided only by using boundary layer control. In this survey, all diffusers with an equivalent cone angle of greater than 6° are taken as wide-angle diffusers.

A wide-angle diffuser should be regarded as a means of reducing the length of a diffuser of given area ratio, rather than a device to effect pressure recovery. Uniformity and steadiness of the flow at the diffuser exit are of prime concern since this affects the performance of vital components downstream. Effectively, the static pressure rise is sacrificed for compactness and for uniformity and steadiness of the exit flow.

One of the most popular means of preventing separation in wide-angle diffusers is by introducing screens of perforated metal or, more usually, woven wire gauze. For the smaller blower tunnels, the cheaper plastic (nylon or polyester) mesh screens may be strong enough. A screen is usually specified by the "mesh", defined as the number of openings per lineal inch ($1/\ell$), and by the wire diameter (d). Given these two specifications , $\beta[\equiv(1 - d/\ell)^2]$, the screen open-area ratio can be determined.

Gauze screens make the velocity profile more uniform and reduce the boundary layer thickness. The ability of a boundary layer to withstand a given adverse pressure gradient is increased when the boundary layer thickness is decreased. Therefore a screen, besides removing the direct effects of boundary layer growth and incipient separations, gives the boundary layer a new lease of life so that a further pressure rise can be negotiated without separation. However, the corner flow in a wide-angle diffuser of rectangular cross-section is still liable to separate because of secondary flow effects, but this just "fills-up" the corners and does not present a major problem. Screens also reduce the turbulence intensity level in the boundary layer and refract the incident flow, upstream and downstream of the gauze. These last two factors, however, do not necessarily take the boundary layer further away from separation.

The effectiveness of a conventional woven wire screen depends largely on its pressure drop coefficient (K), which is defined as the ratio of the static pressure drop across the screen (Δp) to the dynamic pressure of the uniform parallel flow approaching perpendicular to the screen plane ($q = \frac{1}{2}\rho U^2$). Thus $K = \Delta p/q$; K is a function of β, the Re based on the wire diameter, and the inlet flow angle. The refractive index of the screen (μ), defined as in optics, is about $0.9\sqrt{1+K}$. Allowance for its effect on lateral component velocity (Batchelor, 1970) modifies simple formulae for the attenuation factor for velocity non-uniformity. The simple formulae suggest that a single screen with $K = 2$ eliminates small non-uniformities.

A detailed analysis of flow through gauze screens is given below in Section 3.

In a wide-angle diffuser it is better to use several screens, each with a relatively low value of K (say between 1 and 2), placed preferably at the points where the flow would otherwise

separate, than to use a single screen with a large K. This is because increasing K at one station in the diffuser has little effect on the skin friction at a station much further downstream. The most efficient diffuser, needing the fewest screens, will usually have curved walls (in side view), minimizing the adverse pressure gradient. However, straight-wall diffusers, with discontinuities of wall slope at entry and exit, are easier to build and perform satisfactorily if the equivalent cone angle is not too large. It is worth noting that in inviscid flow a stagnation point occurs at a concave corner and an infinite velocity at a convex corner, and it is only the presence of the boundary layer that smooths out the flow. In the absence of boundary layer control, separation is liable to occur upstream of a concave corner, with reattachment further downstream, thus setting up a separation bubble in the corner. In the case of a convex corner, persistent separation is liable to occur at the corner. The positioning of screens, as will become apparent later, is also of vital importance, particularly in straight-wall diffusers.

Diffuser behaviour is greatly affected by the flow condition at entry. A relatively thin boundary layer with steady flow conditions at entry would no doubt delay separation and improve pressure recovery considerably. Conversely, non-uniform or inclined entry flow can lead to early separation. Liepe (Kachara, 1973) investigated the effect of swirling flow, characteristic of single-inlet centrifugal blowers (see Section 4), in conical diffusers and found that the distribution of longitudinal velocity component was more important than that of the circumferential one. Liepe concluded that although rotation decreased the diffuser efficiency, it was possible to use it to advantage in preventing separation. In our tests on centrifugal blowers (Section 4.2) it was also found that swirling flow could, if controlled, aid flow attachment in a wide-angle diffuser. The effect of inlet conditions on the outlet velocity profile is less in screened diffusers than in any other type of diffuser.

Diffuser efficiency (η) can be defined as the ratio of the gain in pressure energy to the loss in kinetic energy. For the simplest case where the velocity is uniform over the cross-section

$$\eta = \frac{p_{out} - p_{in}}{\frac{1}{2}\rho(U_{in}^2 - U_{out}^2)} \quad . \tag{1}$$

This efficiency is generally low when the prevention of separation depends on the action of one or more screens. The principal losses are due to the pressure drop across the screens. Since this is not of primary importance in blower tunnel entry diffusers, little will be said about efficiency in what follows.

Section 2.2 presents an analysis of about a hundred wide-angle diffusers, installed mainly on low-speed wind tunnels. Section 2.3 reviews further wide-angle diffusers using boundary layer control devices other than screens and Section 2.4 is a discussion of design rules based on the analysis of Section 2.2.

2.2. Optimum design charts for wide-angle diffusers

The four most important parameters in a wide-angle diffuser are:

(i) the area ratio, A;
(ii) the diffuser angle, 2θ;
(iii) the number of screens within the diffuser, n;
(iv) the pressure drop coefficient of each screen, K.

The diffuser angle 2θ is defined as the angle between two straight lines joining the inlet to the outlet, as shown in Fig. 4. If the diffuser angles are different in plan view and in elevation, the *larger* is taken to define 2θ. This definition is slightly different from the "equivalent cone angle" defined in Section 2.1 but is probably more relevant for short diffusers.

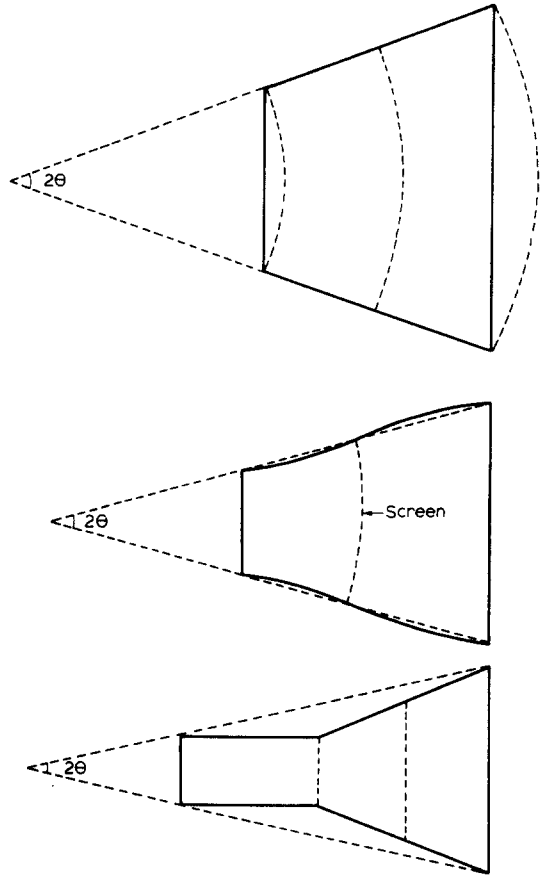


Fig. 4. Typical wide-angle diffusers showing definition of diffuser angle 2θ.

In analysing published data, the pressure drop coefficient of each screen is estimated in terms of β, from Collar's (1939) equation: $K = C\ (1 - \beta)/\beta^2$, where the constant C is taken as 1. This is valid for normal flow at speeds above 10 ms^{-1} and probably over-estimates slightly, but consistently, in typical wide-angle diffusers (see Section 3.1.2).

Schubauer and Spangenberg (1948) inferred design rules based on their own experimental results. In this review we have collected data from many sources. Results are biased towards the Schubauer and Spangenberg rules because many designers have relied on them. The present review can be regarded as a reassessment and extension of the 1948 rules. Schubauer and Spangenberg used screens to obtain "fully filled" conditions (negligible regions of separation) in a straight-walled diffusers of circular cross-section, aiming for zero pressure recovery. They also designed and tested a natural streamline-shape diffuser, and derived a relationship between A and K for straight-walled diffusers (Fig. 7).

The data collected for this review were taken not only from the "successful" diffusers, but also from a few in which separation occurred or which otherwise performed unsatisfactorily.

A "successful" configuration is defined herein as one in which the flow is attached almost everywhere and has a fairly uniform velocity profile with a low turbulence intensity level at the outlet. On the whole, diffusers of which their designers have been proud enough to write them up in the literature have been assumed successful. As well as recording parameters (i) to (iv) above, a note was made of some of the other geometrical parameters of the diffuser, like the cross-sectional and wall shapes, the screen shape (i.e. whether plane or curved), and of the type of wind tunnel (return or open-circuit) to which the particular diffuser was fitted. All the collected data are tabulated in chronological order in Table 1, and combinations of parameters are plotted in Figs. 5, 6 and 7.

It must be noted that most of the diffuser geometry measurements were taken from scaled drawings and are thus susceptible to small errors. It is thought, however that this will not affect the overall conclusions in any critical way.

In Fig. 5, the area ratio is plotted against the diffuser angle. The number of screens for each configuration is specified, and smooth curves have been drawn to enclose the successful configuration points for combination of 0, 1, 2 and 3 screens. The diffuser cross-section and wall shape for each configuration are also shown in this figure. Line AB is a rough boundary between diffusers with straight walls and curved walls and curve CD is the zero screen, unseparated flow limit suggested by Kline *et al.* (1957).

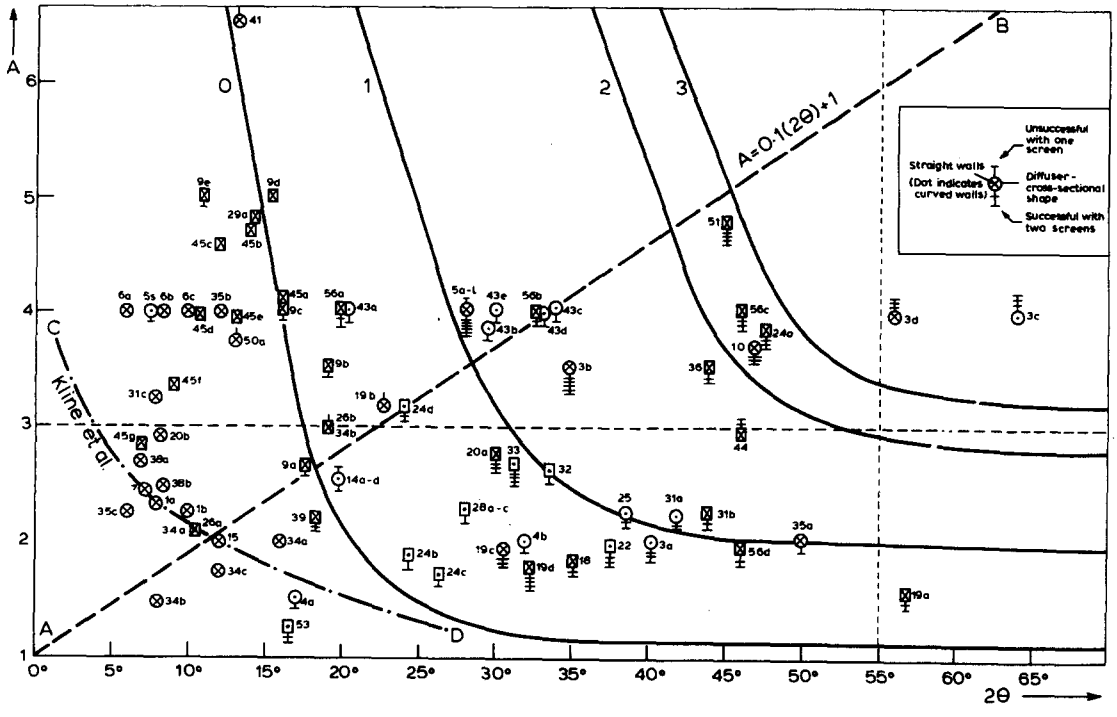


Fig. 5. A vs 2θ for diffusers with screens (or without any boundary layer control device). (Numbers refer to Table 1).

Figure 6 is based on the same axes as Fig. 5, but to a different scale, and shows diffusers using various other methods for boundary layer control.

Figure 7 is a plot of the sum of pressure drop coefficients of all the screens, K_{sum} $(\Sigma \frac{\Delta p}{q})$, vs the area ratio. A straight line (EF) is drawn so as to include the maximum number of successful configurations, and passing through the point: $K_{sum} = 0$, $A = 1$, which represents

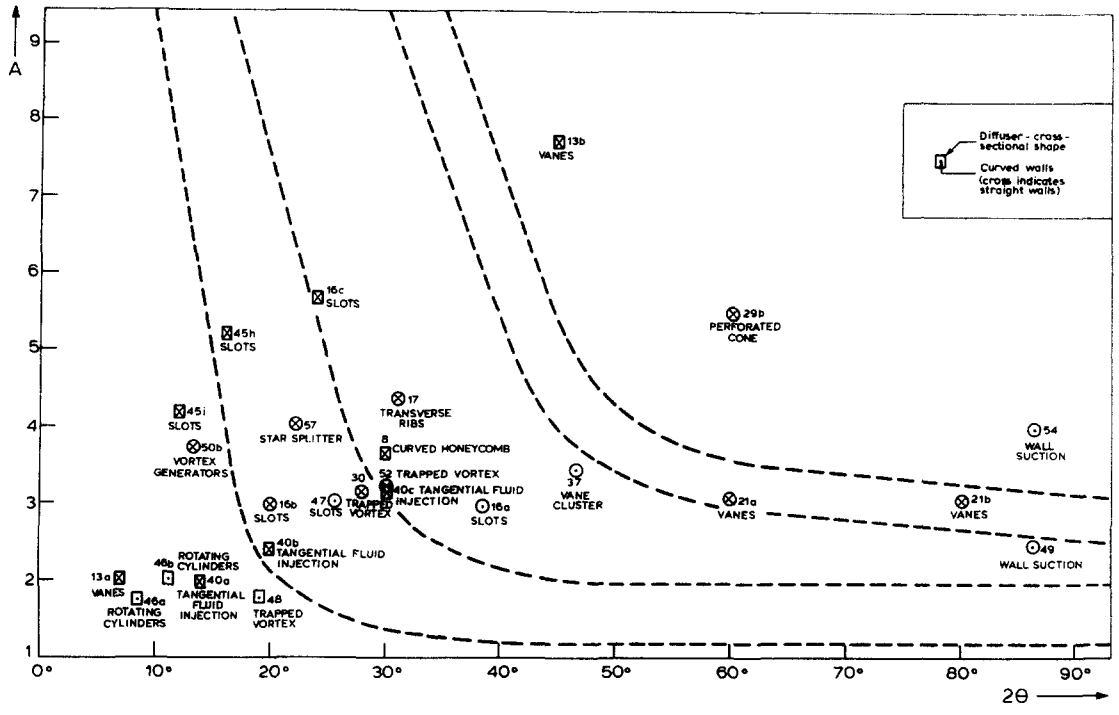


Fig. 6. A vs 2θ for diffusers employing devices other than screens. (Dotted lines are boundaries from Fig. 5).

the limiting case of no diffuser at all. In this figure, diffusers containing curved screens are distinguished from those containing plane ones. A configuration is defined as containing curved screens if *half or more* of the total number of contained screens are curved. The dotted and chain-dotted lines in this figure show the formulae of Schubauer and Spangenberg (1948) and of Squire and Hogg (1944), respectively.

The implications of these results for wide-angle diffuser design are discussed in Section 2.4.

2.3. Other types of boundary layer control devices

2.3.1. Splitter system. The main effect of a splitter system is to increase the total displacement thickness of the boundary layers so that the effective diffusion angle becomes smaller. In potential flow, a symmetrically placed splitter would have no effect.

Splitters in wide-angle diffusers were extensively tested by Raju and Rao (1964). The system consisted of a conical diffuser, segmented into eight, roughly triangular, expanding passages. Assuming uniform or axisymmetric flow at the inlet, each passage captures an equal proportion of the diffuser mass flow. Due to boundary layer growth in the smaller included angle of the inner corners of the passages, a larger reduction of mass flow per unit area occurs along these corners. This, together with the high adverse pressure gradient, causes the flow to separate in the corner, and immediately a long bubble is formed near the diffuser axis, the rear portion containing a well-defined, intense vortex. The bubble deflects the incoming flow outwards and after passing through the narrow "throat", the flow is entrained by the vortex. A small separation trip placed on the cluster apex is sufficient to maintain the vortex-type flow pattern. The entrance loss can be reduced by sharpening the vane leading edges. Flow uniformity depends on the vortex position, which can be suitably varied. In Raju and Rao's work, fairly uniform flow was obtained at the exit, with about 55% efficiency, at an area ratio of 15 (configuration 23 of Table 1).

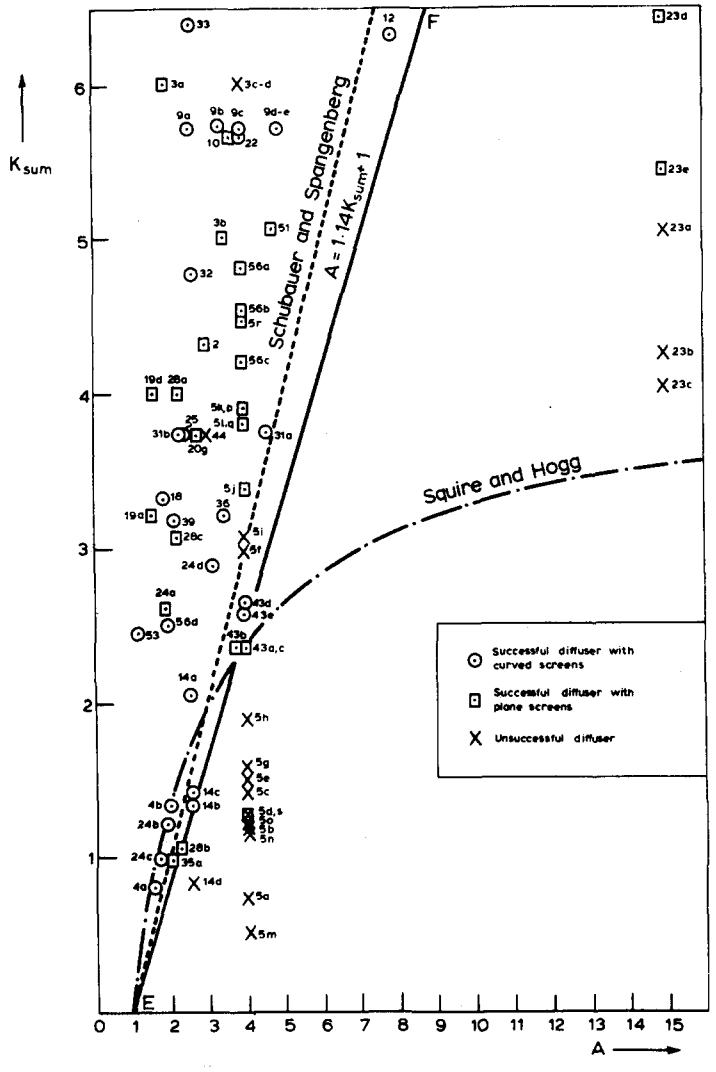


Fig. 7. K_{sum} vs A for diffusers with screens.

Welsh (1976) reviews some passive flow control devices including one with radial spikes—"star", which is a variation of the Raju and Rao idea, although this device has finite thickness and does not span the whole length of the diffuser. He found that a star with eight arms proved most satisfactory; the same number as Raju and Rao, although this system produced a double-peaked profile. The geometry and location of the star were found to be strongly dependent on the inlet flow condition. Welsh successfully used a star flow control device in a wide-angle diffuser of area ratio 4 and $2\theta = 22^{o}$ (configuration 57).

2.3.2. Suction slots. The idea of using suction slots to control boundary layer separation is originally due to Griffith (Griffith and Meredith, 1936). The wall is shaped so that the diffuser has constant high velocity upstream of the slot and constant low velocity downstream. Thus, all the flow retardation takes place over the slot. Let us consider a streamline in the boundary layer, immediately upstream of the slot, above which the fluid possesses sufficient kinetic energy to overcome the pressure increase occurring across the slot. The streamline considered must then be made to stagnate on the downstream slot lip. Thus, by removing through the slot those fluid particles having insufficient kinetic energy to overcome the

adverse pressure gradient, separation is prevented. This category of diffusers, like for example the Griffith diffuser, is found to have a performance far superior to that of comparable straight-walled diffusers. Yang *et al.* (1973) also used slots satisfactorily on Griffith diffusers of both circular and annular cross-section (configuration 47). However, suction slots may not be very successful in diffusers with rectangular cross sections.

A review of boundary layer suction experiments and methods of calculating laminar boundary layers with suction is given in Lachmann (1961).

2.3.3. Trapped vortex. The original idea of separation control by a trapped vortex is due to Ringleb (1961). On the basis of potential flow theory, and using a few results from viscous flow theory he derived the condition for a flow in a 2-D duct containing two standing vortices. The vortex energizes the boundary layer through forced mixing with the outer stream. Difficulty has been experienced in retaining a stable vortex system since energy is lost within the cusp due to skin friction.

There are two common methods of energizing the vortex. Haight and O'Donnell (1974) used a jet from a slot which reinserted into the flow the energy lost (configuration 48). Adkins (1975) used what is probably a simpler scheme, a "bleed-off" from the vortex, which is replaced by air of a higher energy level from the mainstream (configuration 52). By locating a fence just downstream of the vortex, the amount of bleed required for a given area ratio can be controlled. This diffuser exhibited low total pressure loss and high static pressure recovery and so a high efficiency.

2.3.4. Vanes. Small inlet guide vanes placed near or at the diffuser inlet can improve performance appreciably (Moore and Kline, 1958). If the flow is at an incidence to the vanes in "side-view", they act as simple aerofoils, reducing the pressure gradient on the main wall, and this is believed to be their function and mechanism for separation delay. Moore and Kline used multiple, short, straight vanes placed just downstream of the throat and eliminated all separation up to diffuser angles of 45° (configuration 13). For a wide-angle diffuser with $2\theta = 45^{\circ}$ and length to entrance width ratio 8, five vanes (length to entrance width ratio ≈ 2) were successfully used.

A variation of this idea was used by Lawson (1971) for his "egg-box" diffuser. He expanded 3.46:1 from the swirl-free exit annulus of contra-rotating fans to a rectangular duct (height = tip diameter). Three uniformly spaced horizontal plates divided the diffuser into 4 equal area layers. Curved vertical vanes divided each layer into 8 equal-area, S-shaped ducts. The appreciable design and manufacturing effort produced a good outlet distribution with an efficiency of about 0.75.

Another variation of this is to use vortex generators, which shed longitudinal vortices, increasing mixing in the boundary layer. Mabey and Sawyer (1974) prevented separation in their wide-angle diffuser (configuration 50) by mounting two contra-rotating vortex generators of "gothic" planform at 15° incidence. Taylor (1950) gives some useful design criteria for typical diffuser application.

2.3.5. Miscellaneous methods. The Pope and Goin (1965) perforated cone is used in several blow-down supersonic wind tunnels. Their cone faces upstream from the settling chamber, into the diffuser, and spreads the flow through the numerous perforations, although in this respect a downstream facing cone may be more successful. These perforations are designed so that there is a low pressure drop across the cone and a uniform distribution of flow over the duct

area. This type of cone, with an included angle of 90° was used by Azzouz and Pratt (1968) in their inlet diffuser ($A = 5.4$, $2\theta = 60^{\circ}$). Gibbings (1973) used a variation of this method and employed an upstream-pointing pyramid gauze in a wide-angle diffuser of area ratio 16.7 and $2\theta = 54^{\circ}$, a very severe geometry.

Migay (Kachhara, 1973) used transverse ribs, with vortices set up in between, to prevent separation in a wide-angle diffuser. He found that the maximum advantage due to these was realized at $2\theta = 40 - 45^{\circ}$, when the energy spent in the vortices per unit of energy saved due to suppression of separation was a minimum.

Tennant (1973) used a novel form of moving wall for boundary layer control. He had two cylinders, one on each side of the diffuser entrance, rotating in the flow direction. The boundary layer initially formed in the diffuser inlet is accelerated by the moving walls.

2.3.6. Summary of boundary layer control methods. Diffusers with very severe geometries (large A and 2θ) require a large number of screens for satisfactory performance, and there is no reliable method of designing an optimum multiple-screen combination. Access for cleaning becomes difficult and if the entry speed is high, the loading of the first screen may be severe. All these factors make the use of more than about 4 - 5 screens in wide-angle diffusers undesirable. Thus for diffusers with extreme geometries, the other means of boundary layer control discussed above may be worth considering.

All these boundary layer control methods increase mixing in the boundary layer above its natural level and so promote transport of momentum towards the wall, thus delaying separation. The effectiveness of these methods therefore falls with increase in entry boundary-layer thickness. These methods do have certain advantages over screens in that they can cope with solid particles or vapour that might be present in the stream and also do not support any large aerodynamic loads. Furthermore, vanes and splitter clusters can remove any swirl in the flow and can act as sound absorbers or heat exchangers.

However, for common diffuser geometries, screens should be used, since they serve two purposes by not only preventing or delaying separation, but also lowering the turbulence intensity level and steadying the flow. Most of the other devices such as splitters and vanes actually increase the turbulence level by shedding turbulent wakes. Moreover, screens are easier to install than most of the other boundary layer control devices discussed. The results of this survey (Figs. 5 and 6) seem to justify these suggestions.

2.4. Discussion

2.4.1. Design rules. In Fig. 5, the curves enclose diffusers which are claimed or implied by their originators to be virtually free of separation, with fairly uniform velocity profiles and low turbulence levels at exit. A few configurations that did not perform satisfactorily are also shown on the figure and most of them lie outside their respective curves. All the curves have an approximately hyperbolic shape, with two asymptotes. As the value of n increases, the hyperbola vertex moves to a higher value of 2θ and, to a lesser extent, also to a higher value of area ratio.

The asymptotic behaviour of the upper parts of the curves implies that one can have infinitely long diffusers, free of separation. According to Townsend (1976) the maximum diffuser angle for a 2-D channel, for which unidirectional, self-preserving, fully-developed flow is possible without any form of boundary layer control is 4.3°. Thus all the curves should eventually

asymptote towards the line $2\theta = 4.3^{\circ}$, since we are only considering a finite number of screens in the infinitely long diffusers. As far as the lower regions of the curves are concerned, there must exist an accepted separation limit line beyond which, with short and very wide-angled diffusers, we have, in effect, flow over a backward facing step. This maximum diffuser angle probably depends on the pressure-drop coefficient of the corner screen, but a safe limit suggested by the collected data in Fig. 5 is $2\theta = 55^{\circ}$.

The hypothesized limit line (EF) in Fig. 7, which is represented by the equation: $A = 1.14K_{sum} + 1$, includes not only most of the successful combinations, but also a few unsuccessful ones. These unsuccessful configurations satisfying the limiting line in this figure do not in fact satisfy their respective curves in Fig. 5. These particular diffusers have about the right K_{sum} for the given area ratio but too large a diffuser angle. These configurations include 3c and d, and 5f and i. On the other hand, the unsuccessful configuration 14d satisfies its curve in Fig. 5, but not the one in Fig. 7. There are no unsuccessful configurations satisfying both the hypothesized curves except 44 and 50, which will be discussed later.

Thus, an overall design hypothesis is put forward that if a diffuser configuration, with a sensibly-chosen wall shape and sensibly-located screens, satisfies its respective curves in both the figures (5 and 7), then it should perform successfully. Wall shape and screen location are discussed in more detail in Section 2.4.2.

A few of the apparently successful configurations do not satisfy both the hypothesized conditions. Although, as mentioned before, the curves are not inflexible, we now take a closer look at these "abnormal" cases (5s, r; 12 and 23d, e). (Configurations 43 and 56 are used as test-cases and are discussed in section 2.4.3).

Diffuser 12 was used as a jet muffler and thus primary attention was given to reducing the noise level of the jet. The walls were made of perforated metal sheet with a 2 in thick outer covering of fibreglass. This arrangement could, in effect, provide suction at the walls, thus preventing possible separation. The flow did, however, separate at the sharp inlet corner. The other point to note is that this diffuser had a free outlet and thus the adverse static-pressure gradient is not increased just upstream of the outlet, as is the case with diffusers whose wall angle changes abruptly at the start of a parallel duct.

Raju and Rao (1964) tried various screen arrangements for their wide-angle diffuser ($A = 15$, $2\theta = 38^{\circ}$). About nine screen arrangements were found to give relatively steady flow, but only two (23d and e), gave a good degree of filled condition in the diffuser. The exit velocity profiles for these two arrangements were, however found to be very similar to that of a fully developed turbulent pipe flow, implying a high turbulence intensity level, which may not be acceptable. Raju and Rao concluded that a splitter vane cluster would have a better efficiency

In the case of Schubauer and Spangenberg's (1948) diffuser A (5j, k, l), attached flow was achieved in all three cases, but 5j with the six 22-mesh screens gave an overall efficiency nearer to zero, the others giving negative efficiencies. With diffuser B, the suggested optimum combination (5r) of two 54-mesh and one 30-mesh screen gave a slight separation in the diffuser. The five 30-mesh screens (5p), although claimed to be "too many" seem to give a "fully filled" condition and this is the combination suggested for diffuser B by Figs. 5 and 7. The velocity profile for diffuser C (5s) showed that the boundary layer was quite thick and some asymmetry in the flow near the exit was also noted. This reinforces the suggestion of Fig. 7 that a higher K may be needed in this case.

2.4.2. Analysis of the design rules. The rules embodied in Figs. 5 and 7 seem to be a good guideline for wide-angle diffuser design, provided that certain other design factors are kept in mind.

In analysing these factors, it became apparent that diffuser designs fall into two major categories. In the first category are diffusers which are designed just to fulfil the requirements, with the finer design points ignored. In the other category fall diffusers designed with a more cautious attitude, where all the optimum design points have been considered. The latter category includes the larger return-circuit wind tunnels because of the greater financial risks involved.

(i) Inlet conditions. The first important factor involves the flow conditions at entry to the diffuser, which, as discussed above, affect the diffuser performance markedly. In the case of diffuser 50a, it seems surprising that the flow should separate, considering its location in Fig. 5. The reason for malfunction could well be the rather thick boundary layers at the diffuser entry, and a high adverse pressure gradient in the diffuser, due to the high flow Mach number in the working section. On the other hand, Squire (1953) obtained attached flow in a diffuser of area ratio 16 and diffuser angle 10°, thanks to the thin inlet boundary layer.

(ii) Screen positioning. Wall shapes and screen location can be critical. Straight-walled diffusers really need screens at, or just inside, the inlet and outlet to suppress separation altogether, but even this is not essential for acceptable performance, as can be seen from the number of successful straight-walled, one-screen diffusers (Fig. 5). The basic rule, though, is to place a screen where the diffuser wall angle changes, since this is the point where the flow is most likely to separate. The screen positioning (and in fact also the screen shape) is therefore closely related to the wall shape.

A classic example of adverse screen positioning is provided by diffuser 44. This diffuser diverges in two sections, in one plane at a time. The two screens were not installed at the points where the wall angle changed, but halfway along each section. Moreover, a honeycomb was installed just ahead of the diffuser entry, thus straightening the flow which has then to be persuaded to expand through the diffuser—not a recommended design procedure. Thus this diffuser, although it satisfies both the suggested design criteria, failed to perform satisfactorily because some even more basic design factors had not been considered.

Often, in a diffuser without any abrupt changes in wall angle, a screen of relatively high resistance is placed near the entry, followed by one or two, depending on the geometry, of relatively lower resistance near the exit. Schubauer and Spangenberg (1948) showed that with one screen, the extent of separation was reduced by placing it nearer the diffuser entrance. We here propose a more precise theoretical guide to screen positioning.

In diffusers where no obvious screen location is indicated, the following "moment-type" analysis may be applied. We start by defining three non-dimensional parameters:

$$M_o = \sum_{1}^{n} K_r \equiv K_{sum} \tag{2}$$

$$M_1 = \frac{1}{M_o} \sum_{1}^{n} \frac{x_r}{L} K_r \tag{3}$$

$$M_2 = \frac{1}{M_o} \sum_{1}^{n} \left(\frac{x_r}{L} - \frac{1}{2}\right)^2 K_r \qquad (4)$$

x_r denotes the location, measured from the inlet, of a screen with pressure drop coefficient K_r, and L is the overall diffuser length.

M_o or K_{sum} defines the sum of the pressure drop coefficients of all the screens in the diffuser, and for a given area ratio, the minimum acceptable value of M_o is given by Fig. 7. The number of screens required in the diffuser (n) can be determined from Fig. 5. For a given diffuser, having determined M_o and n, the values of M_1 and M_2 have to be optimized by inserting suitable values of x_r/L in eqs. 3 and 4, respectively.

M_1 tends to zero if the screen positioning is biased towards the inlet and tends to unity if the screens are located nearer the outlet. The optimum value of M_1, for uniform screen distribution, is about 0.5. The optimum value of M_2, which is related to the spacing between the screens, depends on the diffuser wall angle at inlet and outlet. Sharp inlet and outlet corners require screens if separation is to be avoided there. In this case, the optimum value of M_2 is about 0.17. On the other hand, if the diffuser has curved fairings at the corners (and the use of these is strongly recommended) then the optimum value of M_2 is about 0.04 (lower for a single screen). These values quoted for M_2 are essentially for $n \leq 4$.

Although these M-values should not be taken as infallible, they should be of use as a more concrete guide on screen positioning.

(iii) Wall shape. Although straight-walled diffusers are easier and cheaper to construct, curved-walled diffusers can be used to advantage. The number of screens required in a diffuser employing curved walls could well be reduced, and the efficiency increased at the same time. It is evident from Fig. 5 that almost all the curved-wall diffusers lie to the right of line AB, given by the equation: $A = 0.1\,(2\theta) + 1$, thus implying that curved walls have been used on diffusers with the larger diffuser angles.

Gibson (1910) and Ackeret (Patterson, 1938) found that the diffuser efficiency was improved when the walls were bell-shaped, although large losses were experienced in the exit sections due to the high adverse pressure gradients imposed there. Küchemann and Weber (1953) describe a trumpet-shaped diffuser with a resistance at the outlet. This type of diffuser was found to operate without separation and at the same time, pressure was also recovered.

Earlier, McLellan and Nichols (1942) conducted experiments on this type of trumpet diffuser and obtained satisfactory performance for $2\theta < 30^{\circ}$ and $A < 3$.

For this type of arrangement, the relation $A = \sqrt{1+K}$ seems to hold (private correspondence, D. M. Rao).

Squire and Hogg (1944) analysed the flow of a jet through a screen and this led to what is known as a streamline diffuser, with an inflected wall-shape. The desirable condition of constant-pressure diffuser walls would be realized if these were shaped to coincide with the free streamlines of the jet. Pressure recovery would obviously not be possible, but flow separation could be prevented, and, according to Gibson (1910), the performance of such a diffuser might be superior to that of a diffuser with multiple screens, possibly involving an overall pressure loss. The same idea of shaping the walls was also used by Yang *et al.* (1973).

where instead of a screen, suction slots were used at a similar streamwise position. Schubauer and Spangenberg (1948) used this idea to design a successful one-screened diffuser (5s).

Potential flow methods are sometimes used to determine diffuser wall shapes. The common method is to represent the potential flow through the duct on a complex plane and specify the required boundary velocity on a hodograph plane. This is then related to the physical plane by conformal transformation. Kachhara (1973) used this method to design his axisymmetric screened-diffusers and found that, in general, the theoretical and experimental results differed, mainly due to the rapid boundary layer growth. He concluded that potential theory is justified in designing wall shapes for diffusers in which boundary layers do not separate, but remain thin along the entire length. These necessary conditions reflect the inadequacy of potential flow methods in simulating real flow situations. However, it is perhaps fair to say that potential flow methods are good enough in most cases, although it may be just as easy to "eye-design" wall shapes and then subject them to a trial and error treatment.

Very often, though, straight-walled diffusers are employed, with curved screens installed at the points of discontinuity in wall angle. Sharp inlet corners must, however be avoided since they undoubtedly promote separation. The need for a radiused inlet was re-emphasized by the results of some experimental work done recently at Imperial College (Section 2.4.3). In the simplest case, the fairing need merely be a bent piece of plywood, tacked over the sharp corner.

(iv) Screen shape. It is an advantage for the screens to intersect the diverging walls and streamlines at right angles, so that the refraction of the flow by the screens does not itself tend to produce separation (see Section 3.1.2). So curved screens, usually in the form of circular arcs, are installed in the diffuser, usually where the wall angle changes, at right angles to the downstream wall so as to refract the flow in a desirable direction. Curved screens can be held in metal frames (e.g. channel sections) pressed into circular arcs and lined with wooden strips so that the gauze may be firmly embedded between two frames. In 2-D diffusers, the screens may be clamped on two sides only, with plenty of slack. It has been found (Bradshaw, 1972) that at practical tunnel speeds the screens fill out into a nominal catenary shape with no apparent tendency to unsteadiness. Small gaps on the unsupported sides do not seem to matter: in fact they probably help in evening out the boundary layer. On inspecting the collected data, it was found that in most cases curved screens had been used when the walls were curved as well.

Another alternative is to use a plane, "variable-K" screen. This normally comprises one screen, concentrically superimposed on another (with the same K-value). This combination then has a higher resistance in the centre and it thus imposes a higher pressure drop on the relatively higher speed regions near the diffuser axis. Kachara (1973) found that for equivalent values of K, outlet profiles were more uniform with a plane "variable-K" gauze than with a plane "constant-K" gauze. He gives an optimum value, for uniform outlet profile, for the ratio of the screen diameters as 0.84. Of course, the losses through a "variable-K" gauze are greater due to the relatively higher K in the middle.

(v) Cross-sectional shape. Most of the diffusers plotted in Fig. 5, and especially those with an area ratio of over 3, are seen to have square or rectangular cross-sections. The main, obvious reason for this is that rectangular diffusers are much easier and cheaper to build than circular ones. It is advisable to fillet the corners between adjacent walls, especially in small tunnels, whose designs are likely to be more adventurous. Gibson (1910) considered

the effects of three sectional shapes, circular, rectangular and square, on diffuser efficiency. He concluded that for a given rate of expansion, the circular duct was the most efficient, followed by the square one. As mentioned before, for wide-angle diffusers used as entry diffusers, as most of the ones plotted are, pressure recovery is not of major importance, and this probably explains the preference for square cross-sections in practice. So unless pressure recovery is important, a rectangular or square diffuser cross-section (with corners filleted) is adequate.

(vi) Tunnel circuit. In Table 1, a note is made of the type of tunnel circuit from which the particular diffuser configuration is taken, and most are found to originate from open-circuit tunnels. This is because most of the return-circuit wind tunnels employ long, gradually expanding diffusers with 2θ of about 5°, so that at most, only a moderate expansion through a wide-angle diffuser is needed. Most wide-angle diffusers, especially those of high area ratio, were designed for small blower tunnels and it is therefore not surprising to see (Table 1) that all the diffusers with $A > 4$ originate from open-circuit tunnels.

2.4.3. Comparison and verification of design rules. The curve (CD) in Fig. 5 is the limiting curve for no appreciable separation, in a 2-D duct with no screens, as suggested by Kline *et al.* (1957). Noting that it is for a 2-D duct, it still seems to suggest configurations of a very conservative nature. The general shape is tending towards that of the newly hypothesized zero screen limit.

In Fig. 7, the second postulated curve for successful operation is compared with two existing theories. For a diffuser with n screens, Schubauer and Spangenberg (1948) derived a formula for a straight-walled diffuser of circular cross-section, which gives the area ratio in terms of K for each screen as the n - th order product:

$$\frac{A_n}{A_{in}} = (K_1 + 1)^{\frac{1}{2}} \cdot (K_2 + 1)^{\frac{1}{2}} \cdot \cdots (K_n + 1)^{\frac{1}{2}} . \qquad (5)$$

Since a diffuser with two screens of equal K is quite a common configuration, this optimum combination line is plotted (dotted in Fig. 7) using this formula, which reduces to: $A = K + 1$ This curve is seen to lie just within the postulated curve ($A = 1.14\ K_{sum} + 1$), thus enclosing most of the successful combinations. On recalculating several of the configurations, using this formula, it was found that they not only lay within the line EF, but very close to the actual values used. This suggests that many designers have relied on the rules put forward by Schubauer and Spangenberg (1948).

Squire and Hogg (1944) made a simplified analysis of screen combinations in relation to wind tunnel design, which yielded some interesting results. They considered an example of a jet in an infinite, ideal fluid impinging normally onto a screen, and using the Bernoulli and continuity equations they derived

$$A = (4 + K)/(4 - K). \qquad (6)$$

This formula holds accurately only for small values of K (less than about 1), since it assumes that the velocity is uniform over the cross-section of the jet at the screen position. The theory becomes less and less exact as K increases. In Fig. 7, it is seen that this formula (chain-dotted) agrees well with the hypothesized curve up to $K_{sum} \simeq 2.4$. Beyond this, the curve diverges rapidly towards the asymptote, $K_{sum} = 4$. These theories, with the different circumstances in mind, seem to agree well with the hypothesized curves.

TABLE 1

Details of Wide-Angle Diffusers Used for Data Correlation

Wide-Angle Diffuser Origin	Diffuser Angle 2θ	Area Ratio A	Wind Tunnel Circuit	Diffuser Wall Shape	Diffuser Cross-Section	Number of Screens or B-L Control Device	Sum of Pressure drop Coefficients K_{sum}
1. Patterson (1938)							
(a) Peters	8°	2.34	-	Straight	Circular	-	-
(b) Gibson	10°	2.25	Open	"	"	-	-
(c) Gibson	10°	9	"	"	"	-	-
(d) Gibson	10°	9	"	"	Rectangular	-	-
2. McLellan,Nichols (1942)	30°	3	"	Curved	Circular	1	4.3
3. Squire,Hogg (1944)							
(a)	40.1°	2	"	Curved	"	3	6
(b)	34.7°	3.5	"	Straight	"	5	5
(c)	64.1°	4	"	Curved	"	3	6 *
(d)	560	4	"	Straight	"	3	6 *
4. Squire,Hughes(1944)							
(a)	17.1°	1.5	"	Curved	"	1 (Curved)	0.8
(b)	31.7°	2.0	"	Curved	"	1 (Curved)	1.33
5. Schubauer, Spangenberg (1948)							
(a) Diffuser A	28°	4	"	Straight	"	1	0.72 *
(b) A	28°	4	"	"	"	1	1.20 *
(c) A	28°	4	"	"	"	1	1.40 *
(d) A	28°	4	"	"	"	1	1.25 *
(e) A	28°	4	"	"	"	1	1.48 *
(f) A	28°	4	"	"	"	1	2.95 *
(g) A	28°	4	"	"	"	1	1.55 *
(h) A	28°	4	"	"	"	1	1.90 *

* Unsuccessful Configuration

TABLE 1 (contd.)

(i) Diffuser A	28°	4	Open	Straight	Circular	1	3.04 *
(j) A	28°	4	"	"	"	6	3.38 **
(k) A	28°	4	"	"	"	5	3.89
(l) A	28°	4	"	"	"	3	3.81
(m) Diffuser B	90°	4	"	"	"	1	0.51 *
(n) B	90°	4	"	"	"	1	1.16 *
(o) B	90°	4	"	"	"	1	1.22 *
(p) B	90°	4	"	"	"	5	3.89
(q) B	90°	4	"	"	"	3	3.81
(r) B	90°	4	"	"	"	3	4.48 **
(s) Diffuser C	7.5°	4	"	Curved	"	1	1.26
6. Squire (1953)							
(a)	6°	4	"	Straight	"	-	-
(b)	8°	4	"	"	"	-	-
(c)	10°	4	"	"	"	-	-
(d)	10°	16	"	"	"	-	-
7. Dryden,Abbott(1949)	7.2°	2.42	Return	"	"	-	-
8. Rhoden (1951)	30° -	3.6	Open	"	Rectangular	1 Curved honeycomb.	-
9. Reid (1953)							
(a)	17.4°	2.63	"	"	Rectangular	1 Curved	5.69
(b)	18.9°	3.5	"	"	"	1 (Curved)	5.69
(c)	15.9°	4	"	"	"	1 (Curved)	5.69
(d)	15.2°	5	"	"	"	1 (Curved)	5.69
(e)	10.7°	5	"	"	"	1 (Curved)	5.69
10.Squire,Winter (1953)	46.5	3.66	Return	"	Octagonal	3	5.64

* Unsuccessful Configuration ** Optimum Configuration

Table 1 (contd)

11.Green,Lilley(1955)	58.5°	17.5	Open	Curved	Circular	3 and Suction	
12.Campbell (1957)	45°	8	"	Straight	"	2 (Curved)	6.3
13.Moore,Kline (1958)							
(a)	7°	2.0	"	"	Rectangular	Vanes	-
(b)	45°	7.7	"	"	"	"	-
14.Gibson (1959)							
(a)	19.8°	2.55	"	Curved	Circ→Octag.	1 (Curved)	2.05
(b)	19.8°	2.55	"	"	"	"	1.36
(c)	19.8°	2.55	"	"	"	"	1.40
(d)	19.8°	2.55	"	"	"	"	0.83 *
15 Robertson, Fraser (1960)	12°	2	"	Straight	Circular	-	-
16 Savage (1960)							
(a)	38.4°	2.95	"	Curved	"	Slots	-
(b)	20°	2.95	"	Straight	"	"	-
(c)	24°	5.65	"	"	Rectangular	"	-
17. Migay(1962-Kachara,1973)	31°	4.35			Circular	Transverse Ribs	-
18. Bradshaw,Hellens (1964)	35°	1.81	Return	"	Rectangular	2 (Curved)	3.3
19.Bradshaw, Pankhurst (1964)							
(a) NPL 36"x27"	56.6°	1.54	Return	"	"	2	3.2
(b) NPL Duplex	22.4°	3.2	Open	"	Circular	-	- *
(c) McKinnon Wood's 13' x 9'	30.6°	1.93	Return	"	Octagonal	3	

* Unsuccessful Configuration

TABLE 1 (contd)

(d) Manchester Univ. Low Turb.	32.2°	1.74	Return	Straight	Rectangular	4	4.0
20.Domboldt, Moszynski(1964)							
(a)	30°	2.75	"	"	"	3	3.72
(b)	8.2°	2.9	"	"	Circular	-	-
21. Feil (1964							
(a)	60°	3.05	Open	"	Circular	Vanes	-
(b)	80°	3.05	"	"	"	"	-
22.NPL 9'x7' (1964)	37.7°	2.0	Return	Curved	Rectangular	3 (2 curved)	5.64
23.Raju, Rao (1964)							
(a)	38°	15	Open	Straight	Circular	4	5.0 *
(b)	38	15	"	"	"	3	4.2 ⁂
(c)	38	15	"	"	"	3	4.0 *
(d)	38	15	"	"	"	3	6.4
(e)	38	15	"	"	"	2	5.4
24. CSIRO, Australia (1965)							
(a)	47.3°	1.92	Return	"	Rectangular	3	2.63
(b)	24.3	1.92	Open	Curved	"	1 (Curved)	1.23
(c)	26.2°	1.71	"	"	"	1 (Curved)	0.99
(d)	24°	3.14	"	"	"	2 (Curved) And Honeycomb	2.87
25. Bradshaw (1967)	38.6°	2.27	"	"	Rectangular	2 (Curved)	3.76

TABLE 1 (contd)

26. Carlson et al (1967)							
(a)	10.5°	2.1	Open	Curved	Rectangular	-	-
(b)	19°	3	"	"	"	-	-
27. Rao (1967)	38°	15	"	Straight	Circular	8 Vane Splitter	-
28. Ali (1968)							
(a)	28°	2.25	Return	Curved	Octag→Rect	1	4.0 **
(b)	28°	2.25	"	"	"	1	1.06
(c)	28°	2.25	"	"	"	1	3.02
29. Azzouz, Pratt (1968)							
(a)	14.3°	4.8	Open	Straight	Rectangular	-	-
(b)	60°	5.44	"	"	Circular	Perforated Cone	-
30.Bachmeier (1968)	28°	3.15	"	"	"	Trapped Vortex	-
31. Bradshaw (1968)							
(a) NPL 15"x10"	41.7°	2.26	"	Curved	Rectangular	2 (Curved)	3.76
(b) NPL 36"x36"	43.7°	2.25	Return	Straight	"	"	3.76
(c) NPL 18" Octagonal	7.7°	3.24	Open	"	Circular	-	-
32. Vogel (1968)	33.7°	2.6	"	Curved	Rectangular	1 (Curved)	4.77
33. Imperial College 24"x2" (1969)	31.2°	2.65	"	"	"	4 (2 Curved)	6.4
34. Johnston, Powars (1969)							
(a)	10.5°	2.1	"	Straight	"	-	-
(b)	19°	3	"	"	"	-	- *

TABLE 1 (contd)

35. NAE 30'x30' V/STOL W/T (1969)							
(a)	50°	2.05	Return	Straight	Circular	1	1
(b)	12°	4.0	"	"	"	-	-
(c)	6°	2.25	"	"	"	-	-
36. Imperial College 6" (1970)	43.7°	3.49	Open	"	Rectangular	2 (Curved)	3.2
37. Lawson (1971)	46.7°	3.46	"	Curved	Circular	Vane Cluster	-
38. Pirrello et al (1971)							
(a) Grumman 26"	7°	2.69	"	Straight	Octag.→Circ.	-	-
(b) Boeing 9' x 9'	8.4°	2.56	"	"	Circular	-	-
39. Bradshaw(1972)	18.2°	2.19	"	"	Rectangular	2 (Curved)	3.19
40. Fielder, Gessner (1972)							
(a)	14°	1.98	"	"	"	Tangential Fluid Injection	-
(b)	20°	2.40	"	"	"	"	-
(c)	30°	3.14	"	"	"	"	-
41. Boldman, Neumann (1973)	13°	6.55	"	"	Circular	-	- *
42 Gibbings (1973)	54°	16.7	"	"	Circ→Rect	Pyramid Gauze	1
43. Kachhara (1973)							
(a) Diffuser A	20.2°	3.99	"	Curved	Circular	1	2.37
(b) Diffuser A1	29.3°	3.86	"	"	"	1	2.37
(c) Diffuser B	33.6°	3.99	"	"	"	1	2.37
(d) Diffuser C	34.8°	3.78	"	"	"	1 (Variable-K)	2.69
(e) Diffuser D	29.5°	4.0	"	"	"	1 (Curved)	2.64

TABLE 1 (contd)

44. Manchester Univ. Blower Tunnel (1973)	46.3°	2.98	Open	Straight	Rectangular	2 (Curved)	3.73 *
45. Runstadler, Dolan (1973)							
(a)	16°	4.09	"	"	"	-	-
(b)	14°	4.68	"	"	"	-	-
(c)	12°	4.57	"	"	"	-	-
(d)	10.5°	3.94	"	"	"	-	-
(e)	13°	3.96	"	"	"	-	-
(f)	9°	3.36	"	"	"	-	-
(g)	7°	2.85	"	"	"	-	-
(h)	16°	5.21	"	"	"	Suction Slots	-
(i)	12°	4.15	"	"	"	"	-
46.Tennant (1973)							
(a)	8.5°	1.75	"	Curved	"	Rotating Cylinders	-
(b)	11.3°	2.0	"	"	"	"	-
47. Yang et al(1973)	25.7°	3.0	"	"	Circular	Slots	-
48. Haight,O'Donell (1974)	19.0°	1.73	"	"	Rectangular	Trapped Vortex	-
49 Juhasz (1974)	86.4°	2.5	"	"	Circular	Wall Suction	-
50. Mabey,Sawyer(1974)							
(a)	13.2°	3.74	Return	Straight	"	-	- *
(b)	13.2°	3.74	"	"	"	Vortex Gens	-

TABLE 1 (contd)

51.Watts et al.(1974)	45°	4.77	Open	Straight	Rectangular	4	5.05
52.Adkins (1975)	30°	3.2	"	"	Circular	Trapped Vortex	-
53.G.A.Slow Speed W/T Salisbury,Australia (1975)	16.5	1.27	"	Curved	Rectangular	2 (curved)	2.46
54.Juhasz (1975)	86.4°	4.0	"	"	Circular	Wall Suction	-
55.Smith (1975)	20°	4.0	"	Straight	"	Trapped Vortex	-
56.Imperial College Test Diffusers (1976)							
(a) I	20°	4	"	"	Rectangular	2	4.8
(b) II	33°	4	"	"	"	2	4.5
(c) III	46°	4	"	"	"	3	4.2
(d) IV	46°	2	"	"	"	2 (1 Curved)	2.5
57. Welsh (1976)	22°	4	"	"	Circular	Star Splitter	-

In order to supplement these design rules, the present author conducted some flow visualization tests, using wool-tufts, on diffusers cut from polystyrene blocks. The details of the diffusers tested, and results obtained are given in Table 2. Four different geometries, chosen so that each satisfied a different limit in Fig. 5, were constructed. Each diffuser was built in 2 - 3 sections so that screens could be sandwiched between sections and the whole arrangement then housed in a wooden box attached to a blower tunnel. All the diffusers had straight walls, with sharp corners between adjacent walls and also at entry and exit. Diffusers I, II and III were found to suffer from severe secondary flow effects in the corners between adjacent walls and were consequently fitted with triangular polystyrene fillets. These three diffusers diverged equally in both planes, but diffuser IV was 2-D and was specially constructed to determine the effectiveness of a curved screen. Diffusers II, III and IV had short parallel ducts attached to their exits and thus the adverse pressure gradient was probably increased just upstream of the diffuser outlet, due to the abrupt change in wall angle. Although the surface finish on the diffuser walls was satisfactory, especially on those cut from fine-grained polystyrene, they were all plastered with Polyfilla^{R}, followed by a coat of emulsion paint, giving a smooth, impervious surface.

The first column in Table 2 gives the diffuser geometry and the *theoretical* minimum number of screens and K_{sum} required, as given by Figs. 5 and 7, respectively. The tests were performed at a Re of about 10^5 with two extreme entry flow conditions: an almost fully developed flow from a long (3m) channel and the flow from a contraction with relatively thin boundary layers. The results presented in Table 2 are, however generalized for all entry flow conditions, in accordance with the design rules. The pressure drop coefficients of the screens were determined using Collar's (1939) equation (8 below) and the M-values using equations 2, 3 and 4.

The combination listed last for each diffuser was found to be the optimum and is the one plotted in Figs. 5 and 7. For diffuser I, it seems evident that a higher M_1 (screens nearer the exit) is required, but the sharp inlet, by inducing separation, makes this optimization difficult. One screen seems quite inadequate for diffuser II and this agrees with Fig. 5. With two screens, the wall flow improves with increasing M_1. Diffuser III, which lies quite near the two-screen limit in Fig. 5 (configuration 56c) seems to operate fairly satisfactorily with two screens. The corner and exit flow condition was, however, improved considerably by installing three screens. The results from diffuser IV exemplify the usefulness of a curved screen in maintaining attached flow.

Kachhara (1973) gives a detailed account of the construction and testing of five axisymmetric, wide-angle diffusers. This report was encountered after the present design rules had been derived and is used here as a test case. The geometrical details of these diffusers are given in Table 1 (configuration 43) and their performance details are given in Table 3, which is based on the same format as Table 2.

All the diffusers were found to operate unsatisfactorily (i.e. with flow separation) without any screens, in agreement with Fig. 5. However, they all performed satisfactorily with one screen, when suitably positioned, and this is not in complete agreement with Fig. 5. All but one configuration (43a) are seen to lie beyond the one screen limit line in Fig. 5, although two of them (43b and e) are not very far beyond. In Fig. 7 diffusers 43d and e are within the successful operation line, EF, whereas diffusers 43a, b and c, although not quite within, lie very close to the boundary. The values of M_1 and M_2 for the optimum combinations in Table 3 are, in general, slightly less than the optimum values quoted in Section 2.4.2. It must be

TABLE 2

Wide-Angle Diffuser Test Results (Configuration 56 in Figs. 5 and 7)

Diffuser Details	n	M_o (K_{sum})	M_1	M_2	OBSERVED FLOW CONDITION.
I A=4 2θ=90° n≯1 K_{sum} ≮ 2.5	1	3	0	0.25	Flow near entry attached - tends to separate towards exit.
	1	3	0.33	0.028	Flow downstream of screen attached but separated on two walls upstream.
	2	4.5	0.11	0.18	Flow tended to separate near exit.
	2	4.8	0.12	0.17	Steady and attached but separation on one wall near exit.
II A=4 2θ= 33° n≯2 K_{sum} ≮ 2.5	1	3	0	0.25	Separation on all walls.
	2	4.5	0.12	0.18	Wall flow attached but separation on fillets towards exit.
	2	4.7	0.25	0.17	Flow unsteady and separated between screens, but attached and steady downstream of second screen.
	2	4.5	0.5	0.03	Flow downstream of both screens attached and steady-not observable near entry.
III A=4 2θ= 46° n≯3 K_{sum} ≮ 2.5	2	4.5	0.17	0.17	Separated on two walls downstream of second screen.
	2	4.7	0.18	0.17	Flow condition near exit improved-less separation
	3	6.2	0.38	0.18	Flow downstream of second and third screens attached and steady.
	3	4.2	0.4	0.16	Wall flow attached throughout-slight separation on fillets upstream of first two screens.
IV A=2 2θ= 46° n≯1 K_{sum} ≯1	1	1.5	0	0.25	Separation immediately downsteam of screen.
	1 (Curved)	1.5	0	0.25	Flow attached immediately downstream of screen-separated near exit.
	2	2.5	0.4	0.25	Flow slightly unsteady and tended to separate, near exit.
	2 (1 Curved)	2.5	0.4	0.25	Steady and attached flow throughout.

TABLE 3

Analysis of Kachhara's (1973) Wide-Angle Diffuser Test Results.
(Configuration 43, in Figs. 5 and 7).

Diffuser Details	n	M_o (K_{sum})	M_1	M_2	OBSERVED FLOW CONDITION
A	1	2.37	0.303	0.039	Transitory separation downstream of screen.
A=3.99	1	3.61	0.376	0.015	Separation upstream of screen.
$2\theta=20.2^\circ$	1	2.37	0.317	0.033	Steady attached flow.
$n<1$	1	2.41	0.324	0.031	Steady attached flow.
$K_{sum}<2.62$	1	2.37	0.345	0.024	Steady attached flow.
A1	1	2.37	0.475	0.0005	Separation downstream of screen.
A=3.86	1	3.61	0.565	0.004	Separation upstream of screen.
$2\theta= 29.3^\circ$	1	2.37	0.519	0.0004	Steady attached flow.
$n<2$					
$K_{sum}< 2.5$					
B	1	3.03	0.468	0.001	Separation downstream of screen.
A=3.99	1	3.61	0.534	0.0012	Separation upstream of screen.
$2\theta= 33.6^\circ$	1 (Curved)	2.41	0.494	0.00003	Steady attached flow.
$n<2$					
$K_{sum}<2.62$	1	2.37	0.494	0.00003	Steady attached flow.
C	1	2.37	0.366	0.018	Separation downstream of screen.
A=3.78	1	2.37	0.384	0.014	Separation downstream of screen.
$2\theta= 34.8^\circ$	1	3.03	0.421	0.006	Separation upstream of screen.
$n<2$	1	3.03	0.384	0.014	Steady attached flow.
$K_{sum}<2.44$	1 (Variable -K)	2.69	0.369	0.017	Steady attached flow-uniform outlet profile
D	1	2.12	0.389	0.0122	Separation downstream of screen.
A=4	1	2.37	0.389	0.0122	Separation downstream of screen.
$2\theta= 29.5^\circ$ $n < 2$	1 (Curved)	2.64	0.389	0.0122	Steady attached flow.
$K_{sum} < 2.63$					

noted that Kachhara's (1973) diffuser tests were performed with favourable inlet flow conditions—a uniform velocity profile with thin boundary layers.

Kachhara's results seem to lend support to the design rules, but our test results seem to imply that small separations or unsteadiness may be acceptable in a wind tunnel diffuser. Also, it is clear that these rules must not be taken as infallible or inflexible, but it is hoped that they will be of use as a confirmation and extension of Schubauer and Spangenberg's (1948) work.

3. SCREENS AND HONEYCOMBS

3.1. Screens

3.1.1. Basic principles. Screens are normally made of metal wires (e.g. brass, phosphor bronze and stainless steel) interwoven to form square or rectangular meshes, and are usually installed in the settling chamber to improve flow uniformity throughout the cross-section. Screens in a wide-angle diffuser can, if carefully installed, perform this function as well, besides their specific task of suppressing separation, so that in principle the number of screens in the settling chamber can be reduced.

Wire screens with the open-area ratios normally required (of the order of 0.6) are available commercially and although there is a restriction on the maximum width available in a single piece (about 1.5 m), this does not present a problem for screen sizes normally required for blower tunnels. Screens woven from nylon or polyester threads are also sometimes employed when the wind loads are not expected to be very high. It must be noted that the aerodynamic properties of a plastic screen may differ slightly from those of a metal screen because of the different elastic moduli of the materials, leading to different weaving properties, although there is no direct evidence for this. Plastic screens may even be more uniform.

Gauze screens have three main effects on the flow passing through them:

(i) reduction of mean velocity variation—leading to prevention of, or delay in, boundary layer separation;
(ii) reduction of turbulent fluctuations;
(iii) refraction of inclined flow—towards the local normal to the screen.

The action of the gauze is described in terms of two parameters: K, the screen pressure drop coefficient and α, the deflection coefficient, defined as the ratio of the emergence angle (ϕ) to the incidence angle (θ). θ and ϕ are measured from the normal to the screen as shown in Fig. 8. Note that $\alpha = 1/\mu$, where μ is the refractive index of the screen. For a plane screen placed normal to the axis of the flow, and for $K > 0.7$, Dryden and Schubauer (Taylor and Batchelor, 1949) gave a semi-empirical relation:

$$\alpha \simeq \frac{1.1}{(1+K)^{\frac{1}{2}}} \tag{7}$$

(The derivation of this equation is explained in Section 3.1.2). Quantitative results for the reduction factors are given in Table 4 and are discussed below.

Screens have been used to improve flow quality in wind tunnels since the 1930s. Prandtl (1933) noted that when screens were installed the velocity distribution improved, and he gave a simple theory for this effect. The effect of screens in reducing the turbulence intensity

levels became apparent later. Dryden and Schubauer (1947) gave a physical explanation for this effect: turbulent eddies of larger scale than the screen mesh size are decreased, by much the same mechanism as for mean velocity variation, and although at the same time smaller scale eddies, of the order of the screen mesh size, may be introduced, they decay much more rapidly than the large scale motions. The overall effect is that at some distance downstream both the turbulence intensity and scale have been reduced.

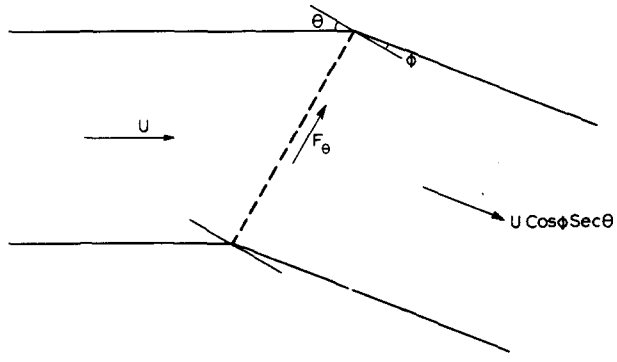


Fig. 8. Flow through an inclined plane screen.

Taylor and Batchelor (1949) produced a detailed analysis of the effect of gauze screens on small disturbances. This theory is linearized on the assumption that there is negligible natural decay of turbulence while the field is translated through the "region of influence" (length comparable to the size of the energy containing eddies). For the case in which the turbulence far upstream is isotropic, the equations for the factors of reduction of turbulence intensity become relatively easy to compute (Batchelor, 1970). The numerical reduction factors given by these equations and Townsend's (1951) measurements (Table 4) are for initially-isotropic turbulence. Dryden and Schubauer (1947) derived a simple theory for the reduction of turbulence intensity based on the assumption that the effect of a screen is partly to absorb the kinetic energy of the turbulence. Both these theories ignore the turbulence set up by the screen itself.

The expression for the factor of reduction of U-component mean velocity (Table 4) implies the interesting possibility that the longitudinal component is completely suppressed by a gauze with $K = 1 + \frac{1}{\alpha}$, which corresponds to $K \simeq 2.8$ if α and K are related as in eq. 7. Also, this expression reduces to the one derived by Prandtl (1933): $1/(1+K)$, when $\alpha = 0$ and to the one due to Collar (1939): $(2-K)/(2+K)$, when $\alpha = 1$. Taylor and Batchelor (1949) show how the analyses leading to these expressions include unrealistic assumptions.

According to the Taylor - Batchelor theory, the lateral-component r.m.s. intensity reduces monotonically with increasing K while the longitudinal component has a flat minimum at $K \simeq 4$. Since, for a given value of K, the reduction of the longitudinal component is greater than that of the lateral component, a reduction of the latter to an acceptable level ensures an acceptable longitudinal intensity. Dryden and Schubauer (1947) give the reduction factor for the turbulence intensity by n identical screens in series as: $1/(1 + K)^{n/2}$. It is thus apparent that for a given pressure loss, a greater reduction in turbulence intensity is obtained by using multiple screens, each with a comparatively low K. Low K is also desirable for other reasons, as will now be shown.

TABLE 4

Factors of Reduction of Mean Velocity Variation and Turbulence Intensity by Gauze Screens.

(i) U - component mean velocity

K	1	2	3	4
$\frac{1+\alpha-\alpha K}{1+\alpha+K}$	0.36	0.1	-0.022	-0.086
(Taylor-Batchelor)	with $\alpha = 1.1/\sqrt{1+K}$			

(ii) V - component mean velocity

K	1	2	3	4
α	0.78	0.64	0.55	0.49

(iii) u - component r.m.s. intensity

K	1	2	3	4
Theoretical (Taylor-Batchelor)	0.44	0.21	0.14	0.13
Experimental (Townsend)	0.45	0.36	0.31	0.28
$1/\sqrt{(1+K)}$ (Dryden-Schubauer)	0.71	0.58	0.50	0.45

(iv) v - w component r.m.s. intensity

K	1	2	3	3
Theoretical (Taylor-Batchelor)	0.74	0.59	0.50	0.45
Experimental (Townsend)	0.66	0.54	0.45	0.40
$1/\sqrt{(1+K)}$ (Dryden-Schubauer)	0.71	0.58	0.50	0.45

Morgan (1960) found that in certain cases the flow behind a uniform screen was unstable with both spatial and temporal variation. Bradshaw (1965) found that this effect was mainly confined to screens with open-area ratios of less than about 0.57, which corresponds to K's of more than about 1.6 (at Re $\simeq$ 100). This instability of flow is probably caused by the jets, emerging from the screen pores, coalescing in random patterns. This sort of spatial instability leads to weak longitudinal vortices which are very effective at producing spanwise variation of boundary layer thickness and surface shear stress in a nominally 2-D boundary layer. These variations are of the order of $\pm$ 10 per cent of the mean values and may be unimportant except in boundary layer studies. For $\beta > 0.57$ ($K < 1.6$), approximately, the variations are much smaller.

The major emphasis in this review is on flow through screens installed in wide-angle diffusers. This means that the streamlines near the diffuser walls approach a plane screen at an angle to the normal (Fig. 9). Except for the effect of the streamwise pressure gradient, this situation can be simulated by placing an inclined screen in a large duct of uniform cross-section. This makes flow visualization and measurements more convenient. The effect of this "angle of incidence" on K and α is determined semi-empirically in Section 3.1.2, which also contains details of some experimental work in progress at Imperial College.

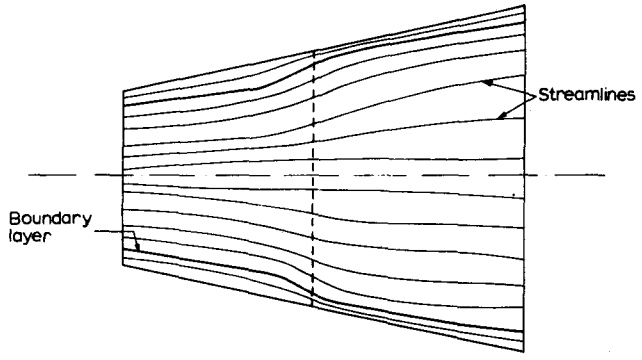


Fig. 9. Flow through a plane screen in a wide-angle diffuser.

3.1.2. Determination of K and α. Over the years, several expressions have been derived giving the pressure drop coefficient of a screen in terms of open-area ratio, β. One of the first deductions and one which is still used is due to Collar (1939). Assuming the resistance of a gauze is proportional to the area of the material and to the square of the speed through the holes, he deduced:

$$K = \frac{C\,(1 - \beta)}{\beta^2} \tag{8}$$

where some dependence of the constant C on Re is to be expected. Collar found that when C was given the value of 0.9, the formula fitted the experimental points (for $U \simeq 9$ m/s) very well. Simmons and Cowdrey (1949) found that for a screen placed normal to the flow, and for velocities greater than about 10 m/s, C = 1.0 gave a satisfactory agreement. The appropriate value of C for these rather higher speeds is therefore probably between 0.9 and 1.0. Note that C is a function of Re and will therefore probably be lower for the typical velocities found in blower tunnel diffusers and settling chambers (5 - 10 m/s).

Wieghardt (1953) plotted K against $Ud/\beta\nu$ (a modified Re based on wire diameter), and found a "best fit" to the experimental data, which included that of Collar (1939) and Simmons and Cowdrey (1949). For $60 > \frac{Ud}{\beta\nu} > 600$, which for typical values of d (0.4 mm) and β (0.5) corresponds to $1.2 > U > 12$ m/s, Wieghardt gives:

$$K = 6.5\,\frac{1-\beta}{\beta^{5/3}}\,\left(\frac{Ud}{\nu}\right)^{-1/3} \tag{9}$$

The effect of the Re factor in eq. 9 diminishes as U increases. The value of K was estimated using Collar's equation (8), with C = 0.9 and Wieghardt's eq. (9) for U = 10 m/s (this being the approximate lower and upper limits for the two formulae, respectively) for typical values of d and β (Pankhurst and Holder, 1952, p. 649). The results are given in Table 5 and although the agreement is not very good, the values are of the same order of magnitude.

TABLE 5

Comparison of Wieghardt's formula with Collar's formula for Determining the Pressure Drop Coefficient of a Screen.

Mesh	Wire Diameter d (mm)	β	K according to Collar's equation 8	K according to Wieghardt's equation 9
7	0.812	0.60	1.00	0.752
8	0.711	0.60	1.00	0.787
9	0.711	0.56	1.263	0.971
10	0.610	0.58	1.124	0.919
12	0.559	0.54	1.42	1.168
13	0.508	0.55	1.338	1.144
14	0.508	0.51	1.695	1.413
16	0.457	0.51	1.695	1.464
18	0.417	0.51	1.695	1.509
20	0.376	0.50	1.80	1.648
22	0.345	0.49	1.912	1.788
24	0.345	0.46	2.297	2.103
26	0.315	0.46	2.297	2.169
28	0.315	0.43	2.774	2.56

So for a screen placed normal to the flow, K may be adequately estimated by Wieghardt's formula for U < 10 m/s and by Collar's formula (with C ≃ 0.9 - 1.0) for U > 10 m/s.

In the case of flow approaching the screen at an angle (Fig. 8), K is found to decrease with increasing inclination (Simmons and Cowdrey, 1949). This is in contrast to intuition which suggests that the effective blockage increases with inclination and thus K must also increase. The actual behaviour is predicted, qualitatively at least, by the analysis due to Taylor and Batchelor (1949), described below.

With the notation of Fig. 8, the force coefficient parallel to the plane of the gauze, F_θ, can be expressed as a function of θ and φ by considering the change in momentum across the screen, giving

$$F_\theta = 2 \cos\theta \sec\phi \sin(\theta-\phi) \ . \tag{10}$$

An empirical relation which holds well in the range $0.7 < K < 4$ is given by Dryden and Schubauer (Taylor and Batchelor, 1949):

$$\frac{F_\theta}{\theta} = 2 - \frac{2.2}{(1+K_\theta)^{\frac{1}{2}}} \ . \tag{11}$$

F_θ is thus a function of K_θ as well as Cos θ. It is therefore hypothesized that the K-formulae should be modified by a factor $\cos^m\theta$ for an inclined flow, with m as an empirical constant. Hence eqs. (9) and (8) become

$$K_\theta = 6.5 \frac{1-\beta}{\beta^{5/3}} \left(\frac{Ud}{\nu}\right)^{-1/3} \mathrm{Cos}^m\theta, \text{ for } U < 10 \text{ m/s}, \tag{12a}$$

and

$$K_\theta = C \frac{1-\beta}{\beta^2} \mathrm{Cos}^m\theta, \text{ for } U > 10 \text{ m/s}. \tag{12b}$$

The only set of comparable data readily available is that from Simmons and Cowdrey (1949), who measured K_θ at various θ (up to 60°) for three gauzes with different β in the speed range 2.4 - 10.4 m/s. For the screen with $\beta = 0.49$, these measured values are compared with those given by eq. (12a) with m = 1, in Fig. 10. The other two screens with $\beta = 0.33$ and 0.39 would not normally be used in wind tunnels. The experimental points seem to agree fairly well with the "theoretical" curves. In the lower speed range, the experimental K_θ seems to increase until a maximum at about $\theta = 10^\circ$, before following the predicted trend. Although Simmons and Cowdrey (1949) offer no explanations for this curious behaviour, it may be attributed to the increased displacement thickness effects at these low Re. Obviously, more experimental data are needed before the value of m can be predicted accurately. However, the general agreement of this equation with experimental data, with m = 1, is promising.

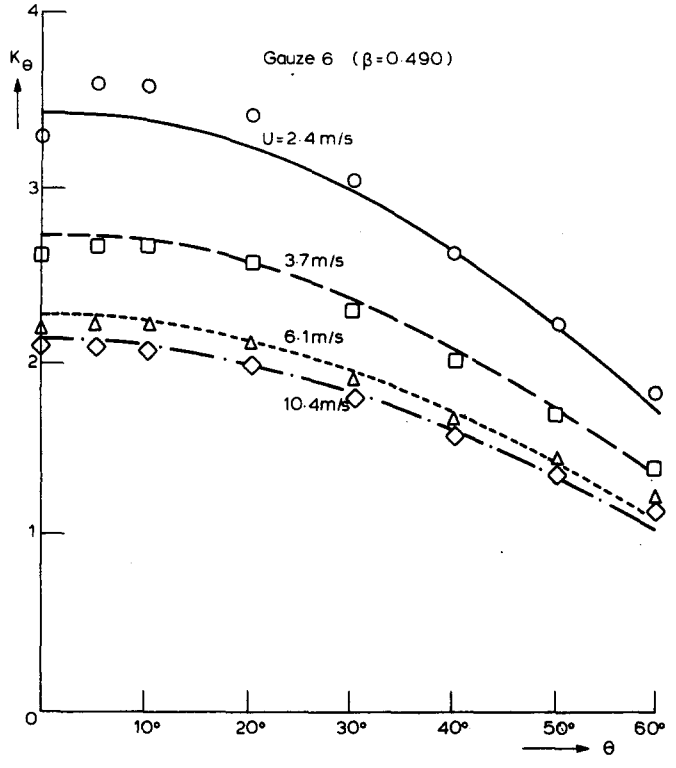


Fig. 10. Comparison of K_θ given by eq. 12a with Simmons and Cowdrey's (1949) results.

Equations (12a and b) have an interesting implication. If we consider flow in a wide-angle diffuser, the angle of incidence of the flow near the walls is roughly half the diffuser angle. The actual angle of incidence *at* the screen is probably larger due to the upstream influence of the screen (Fig. 9). All this suggests that the effective pressure drop coefficient of a screen is less, modified by a factor $\mathrm{Cos}^m\theta$, near the outer edges and thus these low speed regions (e.g. boundary layers) suffer a significantly smaller pressure drop than the high

speed regions near the diffuser axis. This effect probably helps in making the velocity profile more uniform through a plane screen installed in a wide-angle diffuser. This desirable "variable-K" effect would not be realized in a curved screen where the streamlines intersect the screen nearly at right-angles, but of course a plane screen tends to refract the flow towards the axis of the diffuser which is undesirable.

Before formulating α for large θ, we show how eq. (7) is derived for a screen placed normal to the flow. For small θ and ϕ, eq. (10) reduces to:

$$\frac{F_\theta}{\theta} = 2\,(1 - \alpha)\ . \tag{13}$$

Combining eqs. (11) and (13) gives eq. 7: $\alpha = \dfrac{1.1}{(1+K)^{\frac{1}{2}}}$.

Now equating (10) and (11) gives

$$\phi = \tan^{-1}\{\tan\theta - \theta\sec^2\theta\,[1 - 1.1\,(1 + K_\theta)^{-\frac{1}{2}}]\}\ . \tag{14}$$

Hence,

$$\alpha_\theta = \frac{\phi}{\theta} = \frac{1}{\theta}\tan^{-1}\{\tan\theta - \theta\sec^2\theta\,[1 - 1.1\,(1 + K_\theta)^{-\frac{1}{2}}]\}\ . \tag{15}$$

Therefore, given the screen specification (d and ℓ), the stream velocity (U) and the inclination of the screen (θ), K_θ and hence α_θ can be determined from eqs. (12) and (15), respectively.

The only set of satisfactory data available for testing the validity of eq. (15) is in the appendix by Dryden and Schubauer to the paper by Taylor and Batchelor (1949). Simmons and Cowdrey (1949) have a constant ϕ for a given θ but varying stream velocity. Since K_θ is a function of ϕ eq. (14) as well as U (12a), in the speed range investigated, varying U at a given θ must also result in a variation in ϕ. These results therefore seem to be erroneous. Unfortunately, Dryden and Schubauer do not give details of the screens used and so K_θ could not be determined from eq. (12). In the determination of α_θ using eq. (15), the experimental values of K_θ quoted by Dryden and Schubauer have been used. The values of α_θ calculated by this semi-empirical approach are compared with the experimental ones in Table 6. On the whole, the agreement seems quite satisfactory: more experimental data are needed before any final conclusions can be drawn. If we consider the 24-mesh screen at $\theta = 30^\circ$, $(\theta-\phi)$ is about 5°. This means that the flow near the walls in a wide-angle diffuser, on passing through a plane screen normal to the axis tends to deviate from its natural course and towards the axis of the diffuser walls by about 5°, thus opposing the boundary layer effect (Fig. 9) and encouraging separation. In the case of a curved screen, θ would be very small and this undesirable effect would not be realized.

Experimental work is under way at Imperial College on turbulent flow through gauze screens. So far experiments have been conducted on the effect of a plane screen, placed normal to the flow, on a turbulent boundary layer. The screen under test had 24 openings per lineal inch and a wire diameter of 0.254 mm, giving a β of about 0.58 which corresponds to $K \simeq 1.3$ by eq. (8) . The measured value of K was about 1.5.

Velocity profiles were measured at ten stations, almost equally spaced, in a square duct (0.1 m × 0.1 m) with an entry boundary layer thickness of about 5 mm. The most spectacular

TABLE 6

α-values Determined from Equation 15 Compared with the Experimental Values of Dryden and Schubauer.

Angle of Inclination θ	U (ft/sec)	Comparison of α-values for different screens											
		60-Mesh			54-Mesh			40-Mesh			24-Mesh		
		K_θ	α_{exp}	Eqn.15 α	K_θ	α_{exp}	Eqn.15 α	K_θ	α_{exp}	Eqn.15 α	K_θ	α_{exp}	Eqn.15 α
10°	15	4.58	0.42	0.46	2.28	0.68	0.61	1.60	0.69	0.69	0.80	0.84	0.83
10°	25	3.72	0.46	0.51	1.85	0.74	0.66	1.39	0.70	0.72	0.69	0.85	0.86
10°	35	3.40	0.50	0.52	1.65	0.71	0.68	1.25	0.71	0.74	0.64	0.85	0.87
15°	15	4.65	0.43	0.45	2.31	0.61	0.60	1.65	0.63	0.68	0.90	0.85	0.80
15°	25	3.78	0.49	0.49	1.86	0.65	0.65	1.43	0.67	0.71	0.76	0.87	0.84
15°	35	3.44	0.52	0.51	1.66	0.67	0.68	1.31	0.70	0.73	0.72	0.87	0.85
30°	15	3.61	0.45	0.45	1.92	0.60	0.61	1.42	0.68	0.69	0.75	0.82	0.83
30°	25	2.94	0.48	0.50	1.56	0.66	0.67	1.20	0.72	0.73	0.63	0.85	0.87
30°	35	2.64	0.50	0.53	1.39	0.68	0.69	1.09	0.74	0.75	0.58	0.86	0.88
45°	15	2.58	0.48	0.42	1.33	0.64	0.66	1.05	0.72	0.73	0.51	0.84	0.89
45°	25	2.08	0.51	0.51	1.08	0.69	0.72	0.84	0.74	0.79	0.44	0.88	0.90
45°	35	1.84	0.53	0.55	0.99	0.73	0.75	0.74	0.78	0.82	0.41	0.90	0.93

changes in the boundary layer parameters of course take place across the screen and provisional values for these changes are given in Table 7. The "changes across the screen" quoted in this Table are in fact the differences between measurements made at stations at a distance of about 70 mesh lengths (about 2 initial boundary layer thicknesses) upstream and downstream of the screen.

TABLE 7

Experimental Results of Tests on Turbulent Flow Through a Gauze Screen.

Boundary Layer Parameters	Change Across Screen as a Percentage of the Upstream Value.	Recovered value 500-mesh Lengths Downstream as a Percentage of the Upstream Value.
δ_{995} Boundary Layer Thickness	- 50%	60%
δ^* Displacement Thickness	- 75%	55%
θ Momentum Thickness	- 75%	55%
H Shape Parameter	Negligible	-
C_f Skin Friction Coefficient	+ 70%	115%

The concept of a screen reducing the boundary layer thickness and rendering it less susceptible to separation, through increased skin friction, is fully realized from these results. Perhaps the only surprising observation is that H, the shape parameter, does not change significantly across the screen, although the detailed profile shape does.

Future work will include turbulence measurements using hot wire techniques. The investigation of a screen placed normal to the flow is then to be extended to the more interesting case where the screen is inclined. This phase of the work is to include experiments designed to test the validity of the semi-empirical expressions derived above.

3.2. Honeycombs

Screens are not very effective for removing swirl and lateral mean velocity variations. It is preferable, for this purpose, to employ honeycombs, as long as the flow yaw angles are not greater than about 10°. Large yaw angles cause the honeycomb cells to "stall" which reduces their effectiveness. In order to diminish the transverse and swirl components to an acceptable level, and increase the directive effect of a honeycomb, the cell length must be *at least* 6 - 8 times its diameter. The cross-sectional shape of honeycomb cells is usually hexagonal and sometimes square or triangular, mainly for ease in construction.

Honeycombs can also be used for suppressing turbulence (e.g. Loehrke and Nagib, 1976). They tested tubular straw honeycombs made of plastic drinking straws and found that the lateral components of fluctuating velocity were inhibited by the honeycomb cells. Although the honeycombs themselves generate new turbulence, a net suppression of free stream turbulence was achieved by all the honeycombs tested. They observed that the level, structure and decay of the generated turbulence could be modified to advantage by placing a gauze screen at the downstream end of the honeycomb. For reduction of free-stream turbulence they found that the honeycomb cell length should not be *more* than about 10 times its diameter (longer cells result in boundary layer growth and increased turbulence). Besides reducing the turbulence generated due to shear layer instability, this has an added advantage in that the pressure drop across the honeycomb is reduced. However, the resistance coefficient of honeycombs normally used is about 0.5 and this does not contribute appreciably to the tunnel power factor.

In order to obtain the maximum benefit from a honeycomb, it seems that the optimum cell length to diameter ratio should be about 7 - 10. The cross-sectional size and shape of the honeycomb cells are not critical and are often determined by availability. For the suppression of turbulence, however, it is advisable to use small cell sizes; in any case the cell size should be smaller than the smallest lateral wavelength of the velocity variation. In most cases very roughly 150 cells per settling chamber diameter (say a total of 25,000 cells) are adequate.

Impregnated paper honeycombs with cell "diameters" down to about $\frac{1}{2}$ cm are usually adequate when small sizes, less than about 1 m × 1 m, are needed. Any form of deflection or distortion of a honeycomb can change its aerodynamic properties considerably and thus for larger sizes, metal honeycombs must be considered. Aluminium honeycombs like those made for aircraft sandwich construction have more precise dimensions than paper honeycombs and are to be preferred for high performance tunnels. Metal honeycombs can also be made by spot-welding corrugated strips between flat strips, giving a roughly triangular shape, or by sweating together stacks of tubing with hexagonal section ends, as in car radiator manufacture.

With any choice of honeycomb, it is important to ensure that the cells are straight and uniform.

4. CENTRIFUGAL BLOWERS

4.1. Characteristics of centrifugal blowers

Over the past decade, centrifugal fans with "aerofoil-type" impellers have emerged as the optimum choice for driving open-circuit blower tunnels.

The aerofoil-type blades are usually "backward-facing" (Fig. 11)—if forward facing they are less efficient—and in blowers with single inlets, the rotor occupies only about half the width

of the volute casing (Fig. 12). The rotor is asymmetrically positioned so as to accommodate the entry bellmouth within the blower casing, thus reducing the overall width. A bellmouth presents a fairly uniform flow to the blower inlet. The larger blowers tend to have double inlets and the rotor in these is centrally placed, occupying nearly all the width of the casing. For typical blower tunnel speeds and small working section sizes, however, single-inlet blowers are adequate. Besides the rotor, another important component in a centrifugal blower is the splitter plate or "tongue", shown in Fig. 11. The main function of the tongue is to split the flow that emerges from the blower from that which rotates with the rotor.

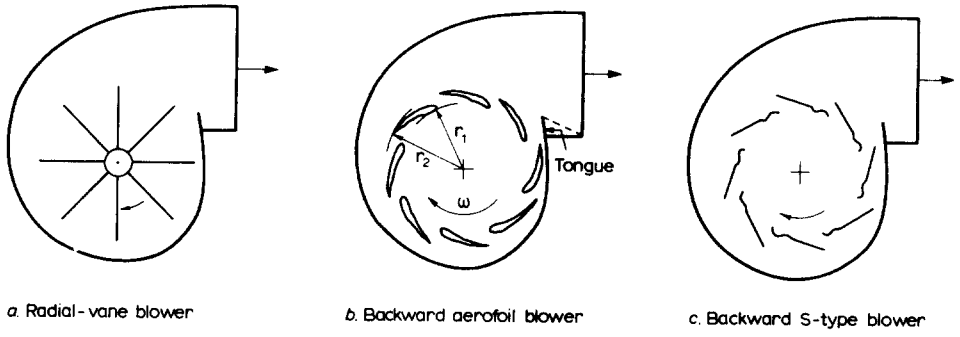


Fig. 11. Different impeller-types used in centrifugal blowers.

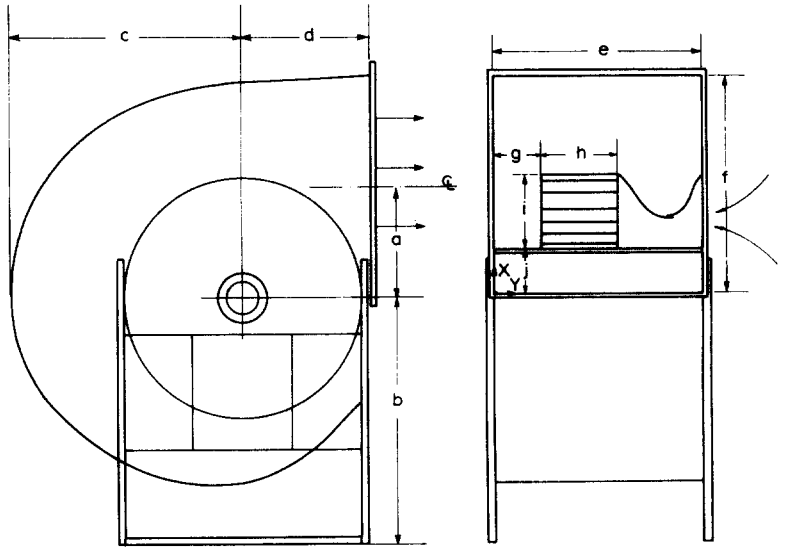


Fig. 12. Centrifugal blower dimensions (lettering refers to Table 8).

Centrifugal blowers are acceptable for high performance tunnels and test rigs, especially when a large variation in power factor is expected. The ability of centrifugal blowers to run with reasonable steadiness and efficiency over a wide range of flow conditions can be attributed to the operation of the whole blading at the same lift coefficient.

The backward-facing, aerofoil-type impeller arrangement has certain advantages over the radial-vane type, which it effectively replaced. Unlike the older type, the aerofoil type blower does not depend on "centrifugal force" to produce the pressure rise. It does, however, depend on the increase in blade peripheral velocity from the inlet to the outlet, which explains the rather curious "backward" blade arrangement. Figure 13 shows the velocity triangle for this sort of blade arrangement. The radial-vane type blower often produced noise and flow

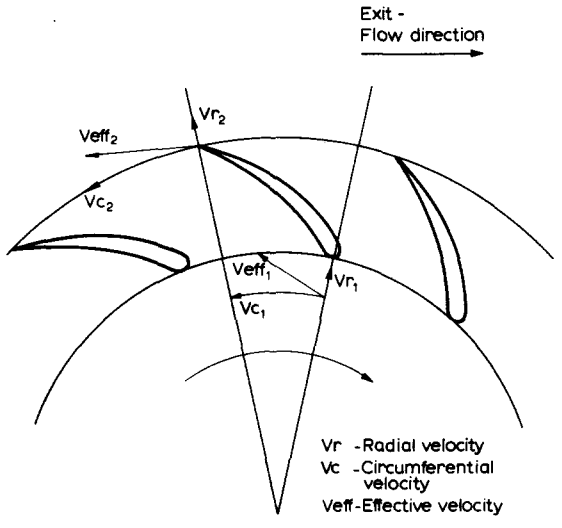


Fig. 13. Velocity triangles for an aerofoil-type rotor.

pulsations which rendered it unsuitable for high performance tunnels. The aerofoil-type blower on the other hand is much quieter.

Since the whole blade area is at the periphery, the pressure rise obtainable at a given peripheral speed and operating condition is greater in an aerofoil-type blower than in an axial-flow fan. In fact, it is also greater than that usually obtained from a radial-vane blower, which is always less than $\frac{\rho(\omega r_2)^2}{2}$ (notation of Fig. 11). Steady and efficient operation can be expected over a greater range of advance ratio ($U/\omega r_2$), than in the case of an axial-flow fan, with a central boss diameter acceptable for wind tunnel use, which stalls first at the roots and produces negative thrust first at the tips. Consequently, the flow uniformity at the outlet of an aerofoil-type blower does not alter as much with advance ratio as does that from an axial-flow fan. Furthermore, the swirl produced by an aerofoil-type blower (Fig. 14) is roughly independent of the advance ratio. All these factors help to reduce the noise and pulsations generated by an aerofoil-type blower to a level well below that of an axial flow fan, particularly at off-design conditions.

The maximum attainable efficiency of an aerofoil-type blower may be as high as 80 per cent, which is competitive with commercial axial-flow fans.

Bradshaw (1967) used a single inlet centrifugal blower with a backward-facing, aerofoil-type impeller on a 15" × 10" (0.38 m × 0.25 m) blower tunnel. Measurements of the frequency spectrum of surface pressure fluctuations in the working section showed no appreciable trace of the blade passage frequency. The velocity-fluctuation or turbulence spectrum showed no trace of either the blade passage frequency or any organ pipe frequency related to tunnel length.

All these favourable characteristics seem to justify the choice of aerofoil-type centrifugal fans for driving blower tunnels, especially since the ability of the blower tunnel to run with the working section exhausting directly to atmosphere means that it can be fitted with a wide range of working sections having a wide range of power-factor contributions, leading to a wide range of blower operating conditions.



Fig. 14. Smoke photograph of the flow emerging from blower B showing only the blower exit area. (The tongue is adjacent to the bottom wall and the rotor is biased towards the r-h side wall).

Although centrifugal blowers are known to produce a satisfactory flow condition in the working section, very little is known about the flow quality at their outlets. It was felt that this was of major importance in relation to wide-angle diffuser performance and thus an experimental investigation, described in Section 4.2, was conducted. Section 4.3 contains a discussion of these results and some suggestions for the choice or design of the optimum blower for a given tunnel configuration.

4.2. An experimental investigation of the aerodynamic properties of centrifugal blowers

During this investigation, four centrifugal type blowers were tested. The dimensional details and test conditions for these blowers are given in Fig. 12 and Table 8. Although all the blowers had single inlets and backward-facing impellers, the impeller size and shape of each was different. The impeller sizes ranged from 15" (0.4 m) on blower B to 27" (0.7 m) on blower C. Blower B and C had conventional aerofoil-type blading, which is perhaps the most efficient and thus the most popular type of blading in centrifugal blowers. Blower D had a simplified version of this, consisting of curved metal plates whereas blower A had what appears to be a new concept of S-shaped blades (Fig. 11).

Since blower A was the one most extensively tested, the results for this are discussed here in detail, with reference to results from the other blowers where relevant. The tests comprised flow visualization, using wool tufts and smoke, and total-pressure scans at the duct exits, with varying loading conditions. Flow visualization results revealed a vortex-type motion emerging from all the blowers (Fig. 14), but the effect of loading on the total pressure distribution could not be generalized. The photographs in this investigation were taken

TABLE 8

Details of Centrifugal Blowers Tested

	BLOWER TYPE	DIMENSIONS (mm)										TEST CONDITIONS			Blower Exit Direction and Specification of Duct
												Zero Loading		Full Load	
		a	b	c	d	e	f	g	h	i	j	Fan rpm	Volume Flow Rate	Operating Point	
A	Keith Blackman 'TORNADO' 20" Impeller 'S' type metal sheet blades.	311	559	502	400	400	622	51	191	152	51	810	1.89 m^3/sec (4,000 ft^3/min)	60	Vertical Exit 0.8 m Parallel Duct
B	Airscrew Fan Centa/Fugal 15" Heba/B type Aerofoil Blades	203	508	356	305	305	406	25	102	89	114	690	0.87 m^3/sec (1,850 ft^3/min)	10	Horizontal Exit 0.3m Diffuser ($2\theta \simeq 19^{\circ}$)
C	Airscrew-Weyroc 27" Heba/B Type Aerofoil Blades	279	838	660	559	559	724	76	165	140	216	1450	7.08 m^3/sec (15,000 ft^3/min)	6	Horizontal Exit 0.5m Parallel Duct
D	Airscrew-Weyroc 21" curved Metal Sheet Blades (Narrow Width)	373	430	530	375	321	479	51	191	152	51	800	1.51 m^3/sec (3,200 ft^3/min)	-	Vertical Exit 0.4m Parallel Duct

in the blower exit plane while all the total pressure measurements were made in the duct exit plane. The plotted total pressures (C_p) have been non-dimensionalized with respect to P_{at} and the blower inlet dynamic pressure.

Figure 15 shows the vortex emerging from blower A, with the centre quite well defined in this wool tuft photograph. This sort of motion was observed from all the blowers—Fig. 14 shows the one from blower B. The reason for this vortex type flow seems to be the asymmetric positioning of the fan (Fig. 12) causing the flow to deflect into the "empty" regions in the volute. The total pressure distribution at the duct outlet to blower A is shown in Fig. 16.

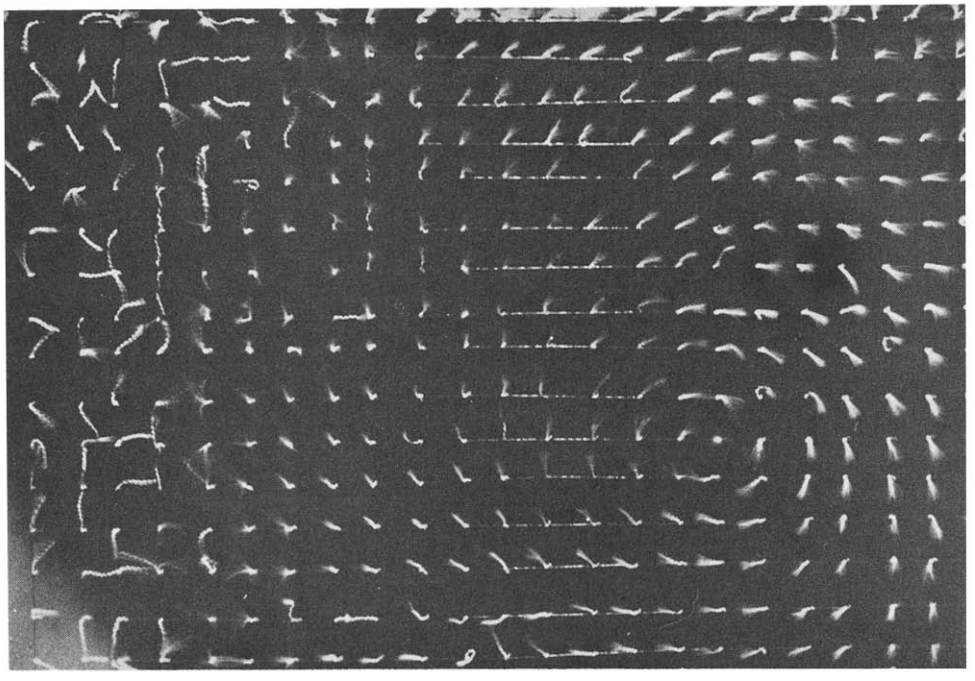


Fig. 15. Wool-tuft photograph of the flow emerging from blower A showing only the blower exit area. (The tongue is adjacent to the 1-h side wall and the rotor is biased towards the top wall).

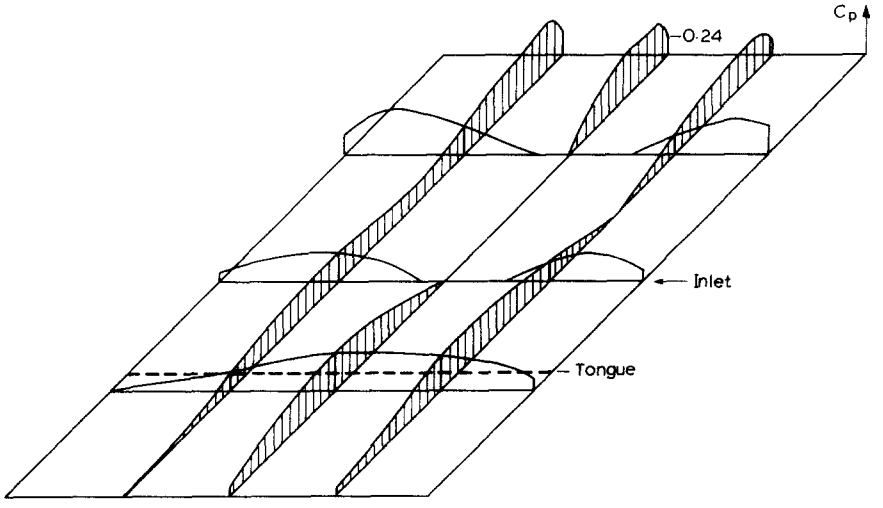


Fig. 16. Total pressure distribution at duct exit: blower A with "no-load".

The flow is seen to emerge from near the sides of the duct, leaving a small region of reversed flow towards the vortex position. In order to assess the flow quality in the absence of a vortex, blower B was partitioned so that the fan was now centrally placed in its half, as in double-entry blowers with full-width rotors. With the vortex almost completely removed, the total pressure distribution was found to be far from uniform (of a parabolic nature). Thus it seems that the vortex might be doing the exit flow more good than harm. The presence of this vortex might improve the uniformity of the blower exit flow, but what is perhaps more important to establish is the effect of this swirling flow on the wide-angle diffuser performance. In order to determine these effects, a wide-angle diffuser ($A = 2.25$, $2\theta = 15^{\circ}$) was attached to blower B. This wide-angle diffuser had straight walls and a rectangular cross-section, and expanded in two sections, in one plane at a time. Total pressure scans were taken at the diffuser exit for three different swirl-strength configurations. For the first test, the swirling flow produced by the blower was fed into the diffuser. The swirl strength was then reduced by inserting two screens in the diffuser: a plane screen was installed at the entry to the first section of the diffuser and a curved screen at the entry to the second section. The pressure drop coefficient of the screens was about 1.5 and thus each screen factored the swirl by about 0.7 (using $\alpha = 1.1/(1+K)^{\frac{1}{2}}$). For the next stage, a paper honeycomb (9.5 mm cell) was installed at the blower outlet and this eliminated almost all signs of the vortex. Figures 17a and b show the total pressure traverses at the diffuser outlet along the centre line in each plane. These results are discussed in Section 4.3.

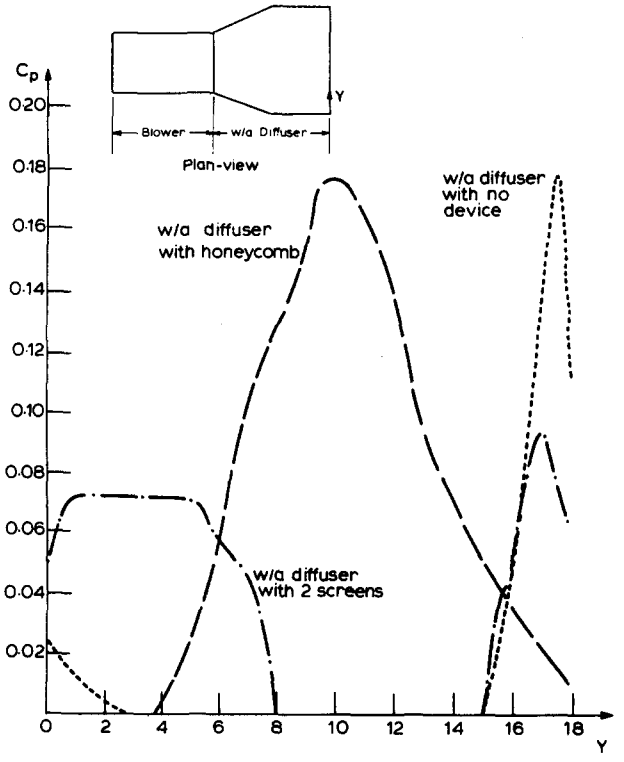


Fig. 17a. Horizontal total pressure distribution at the wide-angle diffuser exit: blower B with varying degrees of swirl.

As the loading on blower A was increased by imposing a static pressure drop through screens at entry, the uniformity of the total pressure distribution was, surprisingly, found to improve. Figure 18 shows the total pressure distribution at the duct exit for this blower at full load. Compared to the no-load case (Fig. 16) the profile is much more uniform, especially in the central areas. The flow uniformity was found to improve even further when a screen

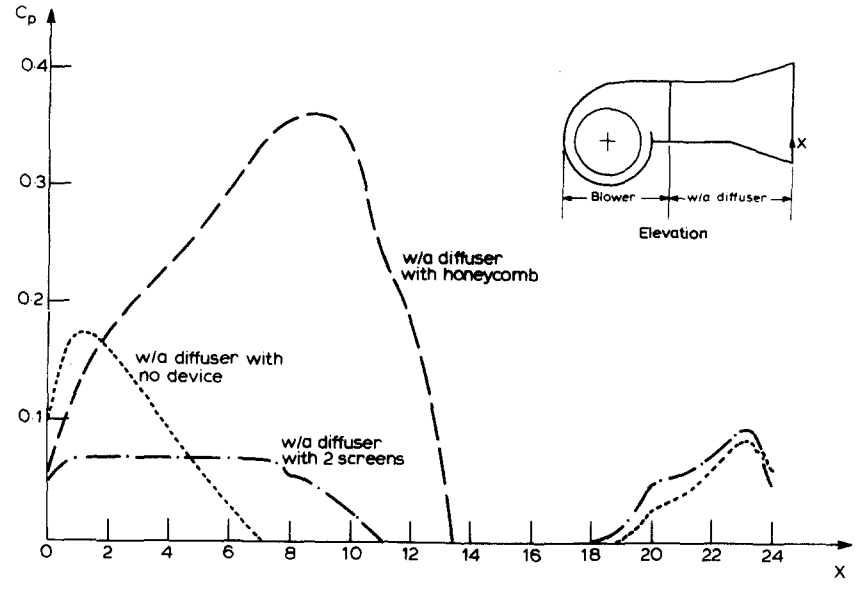


Fig. 17b. Vertical total pressure distribution at the wide-angle diffuser exit: blower B with varying degrees of swirl.

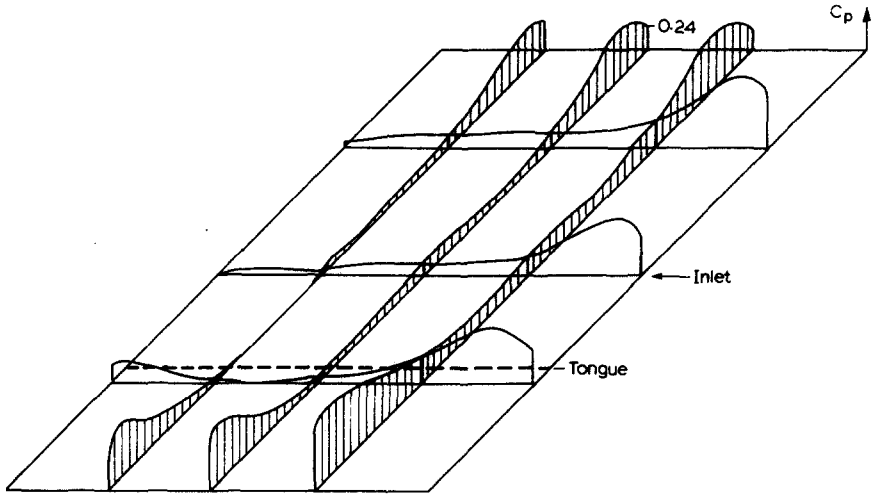


Fig. 18. Total pressure distribution at duct exit: blower A with "full load". Operating point ≡ 460.

(K ∿ 1.5) was installed about a third of the way along the duct. On two other blowers tested for loading effects (B and C), the flow uniformity seems to get worse, or at least does not improve noticeably, as the loading is increased (Figs. 19a and b and 20a and b). This result is perhaps not too surprising. The answer to the unique behaviour of blower A could lie in the impeller type. While blowers B and C have backward-facing, aerofoil-type impellers, blower A has an S-type backward curved impeller. A possible explanation could be that, when loaded, the S-type blades stall relatively early and, through increased mixing, the flow is distributed more evenly at the expense of efficiency.

It was noticed during this investigation that although basically the flow emerged from near the blower or duct side walls, it was more prominent along two adjacent sides. In the case of blower D, this prominence switched-over to the opposite sides when the duct was attached,

but with blower A, no such switching was observed. Total pressure measurements along two opposite duct walls on blower A, in the streamwise direction (Fig. 21) seem to explain this phenomenon. The pressures are seen to vary in anti-phase thus confirming that the orientation of the exit flow is dependent on the duct length: the flow pattern rotates with the vortex.

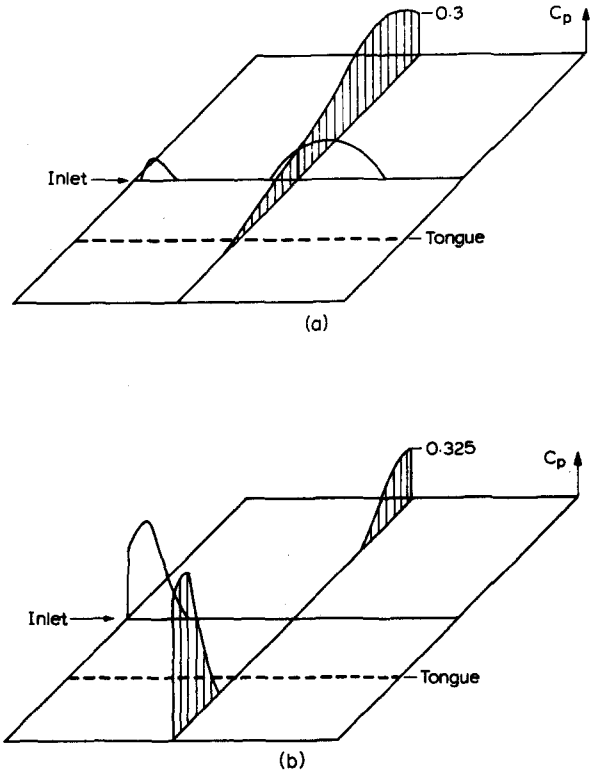


Fig. 19. Total pressure distributions at duct exit: blower B showing effect of loading. Operating point ≡ 10.

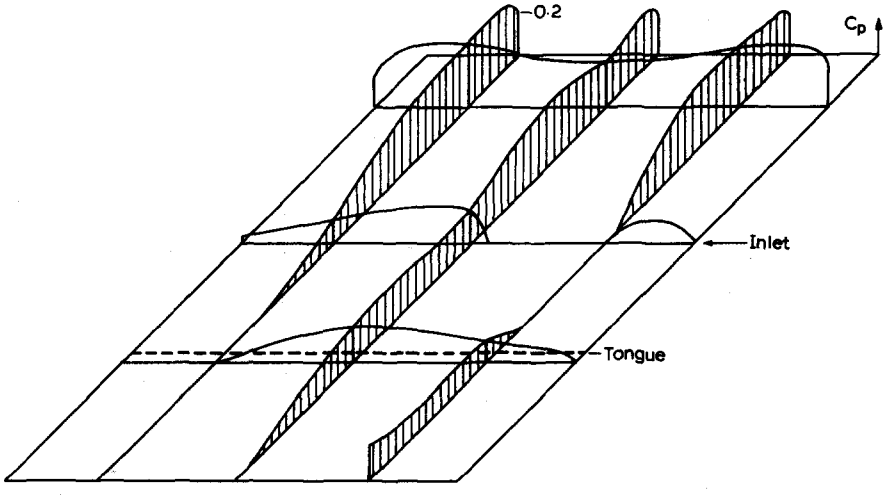


Fig. 20a. Total pressure distribution at duct exit: blower C with "no-load".

These tests show the general (and unique) aerodynamic properties of centrifugal blowers with single inlets. The desirable properties sought are those suited to a blower used at the upstream end of an open-circuit wind tunnel.

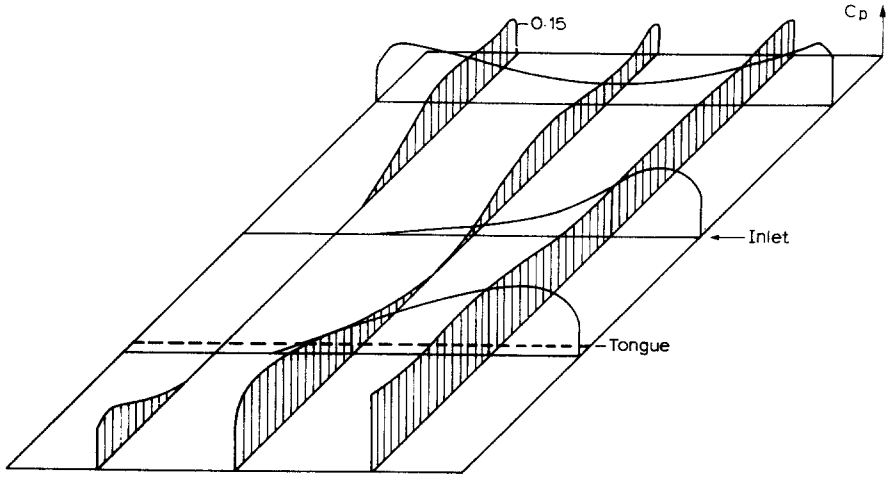


Fig. 20b. Total pressure distribution at duct exit: blower C with loading. Operating point = 5.5.

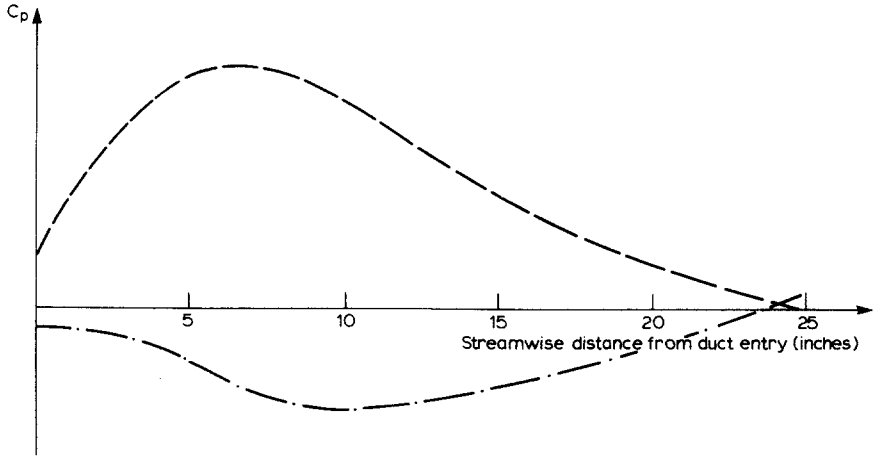


Fig. 21. Total pressure variation in the streamwise direction on opposite walls of duct on blower A.

4.3. Discussion

Two of the four blowers tested (B and C), are attached to blower tunnels at Imperial College. These tunnels have fairly uniform flow profiles (0.2% on total pressure) and low turbulence intensity levels (< 0.1%) in the working section. The flow emerging from these blowers with load is, however, far from uniform (Figs. 19b and 20b). Bradshaw (1967) found that sufficient screening in the wide-angle diffuser and settling chamber eradicated the non-uniformity of the flow emerging from the blower. It must be noted, however, that non-uniform flow from the blower, notably the presence of thick boundary layers, may lead to separation and thence to higher losses and flow unsteadiness in the wide-angle diffuser. These tests thus seem to indicate that although a steady and uniform outlet flow is desirable, departures from it are acceptable, although at an expense in tunnel power factor due to the extra smoothing devices needed.

Although the vortex-type flow is characteristic of all the single-inlet centrifugal blowers tested, the strength of swirl generated by each is probably dependent on the ratio of rotor to blower size.

Figures 17a and b show how this swirling flow, if fed directly into a wide-angle diffuser, can produce an attached flow along the walls but reversed flow in the central regions. With the swirl almost completely eliminated by the honeycomb, the profiles resemble those of a fully developed flow with complete separation on the wall adjacent to the tongue. With the strength of the vortex reduced by the two screens in the wide-angle diffuser, the flow uniformity has somewhat improved and the region of reversed flow is reduced. This is in fact the configuration employed when this blower-diffuser combination is connected to form part of a blower tunnel. It is clear that a swirling component in the flow aids wall flow attachment in the wide-angle diffuser, and by installing screens the uniformity can be improved upon. A swirling component in the blower exit flow may therefore be a property to be sought rather than avoided.

A similar vortex type flow is unlikely to emerge from a double inlet blower, where the rotor is symmetrically placed, though it could produce two small contra-rotating vortices if the ratio of rotor width to casing width is small. One should therefore be more conservative in designing wide-angle diffusers for double-inlet blowers.

Since blowers of many sizes are commercially available, very few tunnel designers will wish to design their own. For a given blower tunnel, once the maximum required fan-static pressure and volume flow rate have been estimated, the makers' performance charts can be consulted. Optimization between the efficiency, RPM and required power leads to the blower choice. For the benefit of designers who may wish to design or modify blowers for specific purposes, some of the critical components are now discussed.

It is essential to have a tongue that splits the flow efficiently since any malfunction in this respect could seriously affect the uniformity of flow in the vertical plane. Besides the size and shape of the tongue, the gap between it and the rotor is also critical. A badly-designed tongue could be improved upon by adding a cusped fairing downstream, as shown dotted in Fig. 11.

Although blade design may influence the flow uniformity at the blower outlet (as seen in Section 4.2), it is certainly less critical than that of axial-flow fans. If blower efficiency is not too important, these blades could be designed in the same way as corner vanes, by choosing a leading edge angle of 4 - 5 degrees and a zero trailing edge angle. The most common blade shape, as on blowers B and C, is that of a cambered aerofoil, with finite thickness. The above tests seem to indicate that S-shaped blades (Fig. 11) although perhaps less efficient, produce exit profiles which are more uniform than those produced by aerofoil-type blades.

The inlet bellmouth can be shaped into a quarter of an ellipse, with its major semi-axis equal to the equivalent inlet diameter and its minor semi-axis equal to three-eighths of the major axis. It is always advisable to attach a filter to the blower inlet so that the screens and honeycomb are protected from foreign particles. Two or more grades of demembraned polyurethane foam, bonded to a non-rusting internal reinforcement, would make a typical inlet filter.

The blower and motor should, if possible, be attached to the same framework. This makes it much easier to manouevre the smaller tunnels, often necessary when more than one blower tunnel is housed in the same laboratory. With the larger blowers, it is advisable to bolt the framework to the floor through anti-vibration mountings so as to reduce resonant effects due to vi-

bration. In this respect, it is helpful to couple the blower to the tunnel with a flexible coupling. Besides reducing vibration, this makes it easier to remove components like screens for cleaning.

The results and suggestions presented in this section have been generalized for most single-inlet centrifugal blowers with backward facing impellers and seem to indicate that, for most purposes, commercially available blowers are adequate.

5. CONTRACTIONS, SETTLING CHAMBERS AND WORKING SECTIONS

5.1. Contractions

Practically all wind tunnels have a contracting nozzle fitted upstream of the working section, for two main reasons: (i) A contraction increases the flow mean velocity and this allows the honeycomb and screens to be placed in the lower speed regions, thus reducing the pressure losses and the tunnel power factor. (ii) Since the total pressure remains constant through the contraction, both mean and fluctuating velocity variations are reduced to a smaller fraction of the average velocity at a given cross section. The most important single parameter in determining these effects is c, the contraction ratio.

The factors of reduction of mean velocity variation and turbulence intensity derived theoretically for axisymmetric contractions, with $c \gg 1$, by Batchelor (1953) are given below:

(i) U-component mean velocity: $\frac{1}{c}$

(ii) V or W-component mean velocity: $\sqrt{c}$

(iii) u-component r.m.s. intensity: $\frac{1}{2c}\{3(\ln 4c^3 - 1)\}^{\frac{1}{2}}$

(iv) v or w-component r.m.s. intensity: $\frac{(3c)^{\frac{1}{2}}}{2}$

(The facter of reduction of *fractional* velocity variation is given by dividing the above expression by C).

It is seen that a contraction is less efficient in suppressing longitudinal turbulence than mean velocity variation. The transverse fluctuations are enhanced, in absolute level, in a contraction: this effect is best explained qualitatively by considering the motion of the vortex elements which constitute the turbulence field. It is sufficient to treat the vortex filaments aligned with and perpendicular to the streamwise direction. The effect of the contraction is to stretch the filament with its axis in the streamwise direction, thus increasing v and w and compressing the filaments perpendicular to the streamwise direction, so decreasing u. The reduction of percentage variation of the lateral component is much less than that of the longitudinal component, so that it is often the desired reduction in the former which decides the contraction ratio. Typically $\sqrt{\overline{v^2}}$ is two or three times $\sqrt{\overline{u^2}}$ after a large contraction.

The design of a contraction of given area ratio centres on the production of a uniform and steady stream at its outlet and requires the avoidance of flow separation in the contraction. Two of the more desirable criteria are: minimum exit boundary layer thickness, and minimum contraction length (which is of course coupled with the space/cost criterion). The danger of boundary layer separation results from the presence, in contractions of finite length, of regions of adverse pressure gradient on the walls at each end. Near the wide end the streamlines are necessarily convex outward, so that the static pressure near the walls is greater than that near the centre line and therefore greater than the average pressure over the cross

section. But the cross-sectional-average pressure at the start of the contraction is closely equal to the pressure far upstream since the cross-sectional area and average velocity are the same. Thus the wall pressure increases from a point far upstream to a point in the wide end, before decreasing as the effect just described is overwhelmed by the effects of decreasing cross-sectional area. The converse applies at the narrow end. (A proof of this for a 2-D-contraction, using the hodograph plane, is given in Bradshaw and Pankhurst, 1964).

It is always possible to avoid separation by increasing the contraction length sufficiently, provided it has a reasonable wall shape. Increasing the length, however, increases boundary layer growth due to skin friction, not to mention the space/cost consideration. On the other hand, a contraction that is too short may suffer from boundary layer separation leading to flow unsteadiness and non-uniformity. It is worth noting that when a contraction is shortened, say by ΔL, the upstream and downstream distances within which local velocity profile distortions are still important on the parallel walls increase, and the effective saving is less than ΔL.

A design satisfying all criteria will be such that separation is just avoided and the exit non-uniformity is equal to the maximum tolerable level for a given application (typically $\pm \frac{1}{2}$ per cent velocity variation outside the boundary layers).

There are certain advantages in using large contraction ratios. The power factor contribution of the screens varies as $\frac{1}{c^2}$ and velocity variations are reduced with increasing c, although Hussain and Ramjee (1976) found that the longitudinal turbulence component increased for c greater than about 4. In particular the effects of small separations at the wide end may be obliterated in contractions with large c. On the other hand, a contraction with a very large area ratio and a reasonable length would have large wall angles which could result in separation near the ends. Also problems due to blower noise could be magnified when large contraction ratios are used because the contraction acts like a horn. Therefore, contraction ratios of between about 6 - 9 are normally used. The upper limit on c is often fixed by the maximum acceptable length of the contraction or the maximum acceptable distance of the tunnel centre line from the floor of the room.

As far as the prevention of separation in a contraction is concerned, one of the important parameters is the cross-sectional shape. In a contraction with an irregular polygonal cross-section the flow near the walls tends to migrate across the corners between adjacent walls. Even if such crossflows are avoided, as in a "two dimensional" contraction or one with a symmetrical cross-section, the boundary layers near the corners will be more liable to separate. The optimum cross-section in this respect is therefore circular. Small contractions can be moulded from fibre-glass, but this is not convenient for large units, and so to compromise between difficulties of construction and undesirable boundary layer effects in the corners, contraction cross-sections in large tunnels are usually octagonal. It is often assumed for design purposes that both the potential flow and the boundary layers behave as in a contraction of circular cross-section. Two-dimensional contractions are not very common, although they present fewer constructional difficulties. Although they require about a 25 per cent longer distance to attain the same wall pressure coefficient and non-uniformity decay rate as an axisymmetric one, they are sometimes preferred on tunnels used for boundary layer studies, where the working section is wide but shallow. Also, the adverse pressure gradients are greater in a two-dimensional contraction and thus a bigger safety margin is necessary.

The contraction wall shape is another important parameter and several theoretical design methods have been developed. In the absence of separation, the flow in a contraction may be adequately described by the Laplace equation. The solution of this linear equation is relatively easy for simple geometries and many analytical solutions have been derived. The solutions for axisymmetric contractions are based on a series solution of either the Laplace equation for the velocity potential (Thwaites, 1946), or the Stokes - Beltrami equation for the stream function (Cohen and Ritchie, 1962), for a given velocity distribution along the centreline (note that the Stokes - Beltrami equation is not quite of the same form as the axisymmetric Laplace equation: see Batchelor, 1967, p. 451). The solution produces an infinite set of stream-surfaces at different distances from the axis, and the most distant one with tolerable pressure gradient (leading to the shortest contraction) is chosen as the wall contour.

The two-dimensional cases are usually solved in the hodograph (U, V) plane. In the hodograph technique, one specifies the relation between the wall velocity $(U^2 + V^2)^{\frac{1}{2}}$ and the wall angle $\tan^{-1}(V/U)$ as a curve in the hodograph (velocity) plane and then transforms this to the real plane. Jordinson's (1961) hodograph method, for large contraction ratios, deserves special mention because he solves the flow in each end of the contraction separately, assuming that, in the limit of high contraction ratios, the shape of the wide end becomes independent of that of the narrow end (and vice versa) because the flow in between is closely radial. This technique has been found to work very well for $c \geq 10$.

Attempts to derive axisymmetric contraction shapes from two-dimensional ones have had limited success. Since most contractions have large contraction ratios, and thus large wall angles, any simple scaling of cross-sectional area is open to large errors. An exception is the approximate method of Whitehead *et al.* (1951) for calculating the flow in an axisymmetric contraction, once the flow in the two-dimensional contraction with the same wall shape is known. Although it is assumed that the streamlines are the same in both cases, the velocity along the streamlines in the axisymmetric case is made to satisfy the axisymmetric continuity equation.

With the onset of computers, numerical methods have now taken precedence. Smith and Pierce (1959) gave a numerical method for calculating the two-dimensional or axisymmetric flow within a given boundary using a distribution of sources on the boundary. More recently, Laine and Harjumäki (1975) computed the potential flow in axisymmetric diffusers by an accurate method in which the Stokes - Beltrami equation is solved with a finite difference approach. Morel (1975) in his numerical study, starts with a given wall shape (formed of two cubic arcs) and solves for the velocities within it. He also plots design charts in which the parameters are: the maximum wall pressure coefficient at the inlet (as in indication of the danger of separation at this end) and at the exit (which is related to the exit velocity non-uniformity). For any choice of these two parameters, the charts yield the "shape parameter" and the nozzle length, for this particular family of shapes.

There is no wholly satisfactory method of theoretical contraction design. Most of the methods, except perhaps that of Morel (1975), are only mathematical tools to be used for the analysis of the flow and offer little concrete design information. The application of all the methods described above require the establishment of some criteria and then the application of trial-and-error techniques, for which limited guidance is given. Also, the methods necessarily simplify the problem by assuming that the velocity profile at inlet is uniform (unlikely in practice).

Because of these drawbacks and because it is often just as easy to choose a boundary shape as it is to choose a suitable criterion, designers have often used the rather unscientific method of design by eye. If this method is employed, it should be noted that the actual form of a contraction contour is not very important except near the inlet and outlet. The wall radii of curvature should be less at the narrow end of the contraction than at the wide end, since a too-gradual transition to the parallel working section delays the development of a uniform velocity profile and this increases the effective contraction length unnecessarily. Also, the radii of curvature at the wide end should be large enough to ensure that the flow through the last screen is not distorted, leading to a non-uniform total-pressure drop. A slowly tapering "infinite-length" contraction can conveniently be terminated at the last screen and the walls made parallel upstream of this, since any extra adverse gradient introduced by the truncation will occur upstream of the screen: again the effect of non-uniformity of pressure drop through the screen must be considered. Ideally, each end must join the parallel sections so smoothly that at least the first and second derivatives of the contour curve are zero (or very small) at the ends. Thus, design as well as constructional care must be taken if a successful contraction is to be produced. Except in small tunnels the boundary layer is likely to become turbulent near the entry, due to the adverse pressure gradient, and if a laminar boundary layer is required at exit (e.g. in jet test rigs), then a rapid contraction must be employed.

Hussain and Ramjee (1976) carried out an investigation on the effects of an axisymmetric contraction shape on incompressible turbulent flow. They analysed four different contraction contour shapes and found that variations within the nozzle, but outside the boundary layers, of the mean velocity and the longitudinal and radial components of turbulence intensity were independent of the contour shape, at least as long as separation did not occur. The boundary layer mean velocity turbulence intensities and thickness were, however found to be distinct functions of the shape. The upstream influence of the nozzle on the mean velocity and turbulence was also found to depend on the contour shape. The nozzle with the cubic equation ($R = a_0 + a_1x + a_2x^2 + a_3x^3$), and the smallest inlet curvature, was regarded as the optimum wall shape. This nozzle gave the smallest boundary layer thickness and the lowest boundary layer turbulence intensitites in the exit plane.

The design methods and analysis described above seem to indicate that a contraction of circular cross-section and a wall shape formed of two smoothly-joined cubic arcs should give optimum results. Theoretical design methods are shown to provide useful guidelines (rather than infallible design techniques) for contraction wall shape design. It seems that in most cases an "eye-design" or a design based on one that is known to be good can serve as well as any other technique, and with less design effort. In either case, it must be remembered that smoothness in contour shape is much more important than dimensional accuracy. Ordinates for two successful "eye-design" contractions are given in Table 9. If the contraction to be designed is of large dimensions and if the design is thought to be critical, then it is advisable to test a model first.

If a model contraction is tested, care must be taken in simulating the exact transition behaviour of the full-size version. It is often most convenient to use a suck-down arrangement for model testing. The model can then be tested with a non-uniform or asymmetric inlet profile, by simply obstructing part of the bellmouth. It is worth ensuring that any small-scale non-uniformities in the full-scale tunnel can be accepted, particularly since the flow out of a centrifugal blower is so non-uniform.

TABLE 9

Contraction Ordinates

a) Two-Dimensional Contraction (c=9) (Bradshaw, 1972)

(Origin of axes on centreline at the upstream end of the contraction)

x	0	2	4	6	8	10
±y	22.50	22.50	22.45	22.40	22.30	22.15
x	12	14	16	18	20	22
±y	21.90	21.50	20.90	20.22	19.50	18.70
x	24	26	28	30	32	34
±y	17.80	16.65	15.30	13.80	12.00	9.00
x	36	38	40	42	44	46
±y	7.50	5.50	4.00	3.07	2.62	2.53
x	47	48				
±y	2.50	2.50				

b) Three-Dimensional Contraction (c=9).

(Origin of axes as for 2-D contraction).

x	0.0000	1.224	2.446	3.666	4.884	6.098
±y	12.00	11.970	11.923	11.856	11.768	11.658
±z	9.00	8.982	8.944	8.954	8.828	8.746
x	7.308	8.511	9.706	10.889	12.057	13.204
±y	11.525	11.362	11.178	10.956	10.699	10.403
±z	8.646	8.524	8.386	8.218	8.026	7.804
x	14.324	15.409	16.446	17.425	18.328	19.137
±y	10.051	9.672	9.232	8.742	8.206	7.627
±z	7.540	7.256	6.926	6.558	6.156	5.722
x	19.832	20.388	20.761	21.126	21.541	
±y	7.073	6.417	5.855	5.455	5.167	
±z	5.270	4.814	4.392	4.092	3.876	
x	21.977	22.425	22.879	23.337	23.798	
±y	4.919	4.767	4.620	4.503	4.404	
±z	3.690	3.576	3.466	3.378	3.304	

TABLE 9 (contd.)

x	24.261	24.725	25.189	25.654	26.120
±y	4.324	4.251	4.204	4.159	4.122
±z	3.244	3.194	3.154	3.120	3.092

x	26.586	27.052	27.518	27.985	28.451
±y	4.092	4.066	4.047	4.031	4.018
±z	3.070	3.050	3.036	3.024	3.014

x	28.918	29.384	29.850
±y	4.007	4.001	4.000
±z	3.006	3.002	3.000

5.2. Settling chambers

In open-circuit blower tunnels, settling chambers are created rather than designed since the frames containing the honeycomb and screens, clamped together, usually form the settling chamber. The usual arrangement consists of a honeycomb (with about 25,000 cells) followed by screens, the number and K-value depending on the turbulence level requirements in the working section. However, more than about two or three screens ($K \sim 1.5$) are not normally required in a well-designed blower tunnel with screens in the wide-angle diffuser.

As mentioned above, a flow with large yaw angles often causes the honeycomb cells to stall thus reducing the effectiveness. Therefore, if severe yaw or swirling flow is expected from the wide-angle diffuser, it is advisable to install one screen upstream of the honeycomb, so that the swirl is reduced. A screen with $K = 1.5$ reduces yaw or swirl angles by a factor of about 0.7. Also, it has been found (Bradshaw, 1972) that spanwise variations of skin friction in the working section may be reduced by placing the honeycomb downstream of the screens. The consequent rise in turbulence level was found to be within acceptable limits.

If the full benefits of a multiple screen combination are to be experienced, then the distance between successive screens should not be less than about 500 diameters. This ensures that the turbulence in the wire wakes of one screen decays before the next screen is reached. The optimum distance between the last screen and the contraction entry has been found (Morel, 1975; Laine and Harjumäki, 1975) to be about 0.2 cross-section diameters. If the distance is much shorter, significant distortion of the flow through the last screen may be expected. On the other hand, if this distance, or for that matter the overall length of the settling chamber, is too long then unnecessary boundary layer growth occurs. However, the settling chamber length could be critical to the tunnel performance if severe separations occur in the wide-angle diffuser.

In small tunnels, screens are normally tacked onto wooden frames so that when clamped tightly together the high spots of the woven-wire screen sink into the frames, giving a firmer hold. The honeycomb is usually just push-fitted into a screen frame. A useful arrangement, for the smaller tunnels, is to rest the wide-angle diffuser, screen frames and contraction on a table

and clamp them by drawbolts, so that frames can be withdrawn easily (e.g. for cleaning and replacing screens). On larger tunnels, it is advisable to equip the settling chamber (and wide-angle diffuser) components with castors for ease of removal.

5.3. Working sections

Working section design is totally dependent on the requirements of each individual experiment. Many blowers have detachable working sections without exit diffusers, a very flexible arrangement. An exit diffuser, the design of which was briefly discussed in Section 2.1, must be fitted if the tunnel power factor is critical. The wide variety of blower tunnel working sections range from quasi-two-dimensional ones (for boundary layer studies or smoke tests) to ones with special features such as slotted walls and flexible roofs. In Fig. 3 some of the many different working section configurations used on the small blower tunnels at Imperial College are shown.

For tunnels which are used for testing models mounted in the working section, the working size and shape are usually dictated by the need to minimize tunnel interference for a predetermined model size. The most popular cross-sectional shape for these working sections is a rectangle of about $\sqrt{2}$ to 1 ratio, with corner fillets. If models like those of aircraft are to be tested, then it is usual for the large (spanwise) dimension to be horizontal, for convenience in measuring forces.

The flow out of a contraction often takes a distance equivalent to about 0.5 diameters before the non-uniformities are reduced below an acceptable level. Also, if a turbulence grid is installed, it may take up to 10 - 15 mesh lengths before a homogeneous flow is obtained. These requirements often limit the minimum length of the working section. On the other hand, a very long working section results in unnecessary boundary layer growth which could lead to separation in an exit diffuser.

The inevitable boundary layer growth in the working section results in a static pressure drop in the axial direction. Any increase in cross-sectional area to compensate for this growth is usually accomplished by tapering the fillets or sometimes by tapering the walls themselves. It is sometimes convenient to obstruct the outlet of the tunnel with a screen to produce an over-pressure in the working section, so that boundary layer control by suction, if necessary, is achieved by merely opening holes.

Most blower tunnel working sections are made of plywood in timber frames. Drumming of plywood panels leads to not only structural fatigue, but also unsteadiness in the working section flow and this must therefore be kept to a minimum, preferably by keeping the panel resonant frequencies above the rotor blade frequency, at maximum speed. It should be the deflection and vibrational properties and not the ultimate strength which decide the material size. Almost all working sections have removable side panels and it is suggested here that these be mounted on pinned hinges. This makes "single-handed" removal of these panels much easier, besides rendering them less susceptible to damage.

A blower tunnel working section without an exit diffuser gives almost as easy access to the model as does the open-jet arrangement sometimes used on return-circuit wind tunnels.

6. CONCLUSIONS

6.1. Wide-angle diffusers

6.1.1. Design rules

(i) Keep to the *left* of the relevant curve and the separation limit line (dotted), in Fig. 5. Small separations (acceptable for tunnel purposes) may still occur.

(ii) Area Ratio must be *less than* $(1.14\ K_{sum} + 1)$, where K_{sum} is the sum of the pressure-drop coefficients of all screens.

6.1.2. Further suggestions

(i) Use screens for boundary layer control if area ratio is less than about 5 and diffuser angle less than about 50°.

(ii) It is safe to use a rectangular cross-section (with corners filleted) unless pressure recovery is important: if so, use circular cross-section.

(iii) Use curved walls if $2\theta > 10\ (A-1)$ degrees.

(iv) Use curved screens to prevent separation.

(v) Avoid sharp corners at inlet and outlet—use fairings.

(vi) For screen location set: $M_1 \simeq 0.5$

$M_2 \simeq 0.17$ (Sharp Inlet)

$\simeq 0.04$ (Radiused Inlet)

(M_1 and M_2 are defined in eqs. 3 and 4, respectively).

6.2. Screens

(i) Use conventional metal wire gauze screens with square (or rectangular) meshes since almost all existing data and theories only apply to these.

(ii) For estimation of K:

(a) Use Wieghardt's (1953) formula for U < 10 m/s.

(b) Use Collar's (1939) formula for U > 10 m/s (with C = 0.9 for the lower speed range and C = 1 for the higher range).

(iii) For a given pressure drop it is better to use multiple screens with relatively lower K

(a) Less than about 1.5 for minimum surface shear stress variation.

(b) About 2.4 for minimum longitudinal mean velocity variation. An optimum value is $K \simeq 2.0$.

(iv) Make sure gauzes are uniformly woven and free of wrinkles.

6.3. Honeycombs

(i) For most purposes impregnated paper honeycombs are adequate.

(ii) For high performance tunnels, or if high loads are expected, use aluminium honeycombs.

(iii) Optimum length-to-diameter ratio of cells is about 7 - 10.

(iv) For suppressing turbulence use small cell "diameters".

(v) Makes sure cells are straight and uniform.

6.4. Blowers

(i) Commercially available, single-inlet centrifugal blowers are adequate for most blower tunnels.

(ii) The level of non-uniformity in the outlet flow of a typical blower with a backward-facing, aerofoil-type impeller is acceptable for most purposes.

(iii) Uniformity of the outlet flow is better with an S-type backward facing impeller—but with a lower efficiency, which probably implies higher outlet turbulence.

(iv) Vortex type flow is characteristic of all single-inlet blowers—and, if reduced using screens, can be used to advantage in the wide-angle diffuser because total pressure is higher near the walls.

(v) It is advisable to connect blower to tunnel with a flexible coupling.

6.5. Contractions

(i) Use contraction ratios between about 6 - 9.

(ii) It is advisable to use axisymmetric cross-sectional shapes for small contractions and octagonal ones for large contractions - two-dimensional contractions may be used for special purposes (e.g. boundary layer studies).

(iii) For wall shape, "eye-design" or an imitation of a successful design is adequate for most purposes. A wall shape contour with a cubic equation also seems to work well.

6.6. Settling chambers

(i) Install the honeycomb ahead of the screens unless swirl angles greater than about 10°—if so, install one screen upstream of the honeycomb.

(ii) Optimum spacing between successive screens is about 500 wire diameters.

(iii) Optimum distance between the last screen and contraction entry is about 0.2 duct "diameters".

6.7. Working sections

(i) Use rectangular cross-section of about $\sqrt{2}$ to 1 ratio, unless for special purposes.

(ii) Control the streamwise pressure gradient by tapering fillets or walls.

(iii) Use an exit diffuser if tunnel power factor is more important than flexibility of working section arrangement.

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